

Stored or Computed? Separating Compositional from Holistic Representations of Binomials in Language Models

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Abstract

Humans balance abstract and item-specific knowledge when ordering binomials: abstract constraints shape the preferences for infrequent binomials, while idiosyncratic knowledge drives them for frequent ones. Evaluating 16 language models across five families, we demonstrate a parallel trade-off. Probing hidden-state representations shows that language models do learn abstract preferences. However, a binomial’s relative frequency in the training corpus predicts a model’s preference over and above anything those abstract preferences supply, and predicts it more strongly for high-frequency binomials. To locate that item-specific component, we zero out cross-word attention, forcing the representation into a strictly additive function of the two words. The intervention mitigates but does not remove the idiosyncrasy of high-frequency binomials, indicating that storage shifts the individual word representations and not only the phrasal one. Examining the developmental trajectory across checkpoints further shows that abstract preferences are present early while the frequency-dependent storage is not observable until later, with the point of detection governed by training tokens rather than model scale.

1 Introduction

Language users are constantly faced with a trade-off between holistically stored, item-specific knowledge and abstract, generative knowledge. For example, the meaning of a phrase often depends on whether it is composed from its individual components or retrieved as a whole. Idioms, such as *kicked the bucket*, are the clearest form of this: the compositional meaning refers to the action of kicking a bucket whereas the idiomatic meaning refers to someone passing away. However, this trade-off isn’t limited to meaning. More generally, the representation of a multi-word expression may be composed from the representations of its individ-

ual pieces, or the expression may be stored and retrieved holistically. How, and crucially when, this trade-off emerges in learning has been a question that linguists have been asking for nearly a century (Houghton, 2025a; Houghton and Kapatsinski, 2026; Houghton and Morgan, 2023, 2024; Morgan and Levy, 2015, 2024).

1.1 Abstract Preferences and Holistic Storage

Historically, these questions have been largely addressed at the behavioral level. Berko (1958) famously demonstrated that children could produce the correct plural of nouns that they had never heard before rather early in life; because the plural form was never available to be stored, it must instead have been generated using abstract knowledge of the language. Evidence for abstract knowledge of this kind has been relatively uncontroversial, and it does not rest on child language alone. The priming literature, for instance, has demonstrated that sentences sharing syntactic structure (e.g., Bock, 1986) or semantic structure (e.g., Meyer and Schvaneveldt, 1971) facilitate the processing of later sentences sharing those features, which requires that the shared structure be represented independently of the particular words that realize it.

Evidence for holistic storage, on the other hand, has been somewhat more controversial. Nonetheless, evidence of holistic storage has been accumulating over the last few decades (Kapatsinski and Radicke, 2009; Stemmer and MacWhinney, 2004). Perhaps the most famous example comes from Bybee and Scheibman (1999). In their seminal study, they demonstrated that the extent to which the word *don’t* is reduced is determined by the frequency of the phrase it occurs in. *I don’t know* can be reduced to a single nasalized vowel, while *don’t* in less frequent phrases, such as *I don’t walk*, is unable to be reduced as much. The context-specificity of this reduction demonstrates that the representation of certain high-frequency phrases

cannot be reduced to the simple composition of the representations of the individual words.

Binomial expressions (e.g., *cats and dogs*) are notable for their relationship between abstract preferences and holistic storage: they show evidence of being driven by both kinds of knowledge. Speakers have strong intuitions about the order of the two words: *salt and pepper* sounds far more natural than *pepper and salt*, even though there’s nothing ungrammatical about the latter. For lower frequency binomials, [Morgan and Levy \(2016\)](#) demonstrated that these preferences are driven by abstract constraints such as the length of a word, the semantic content of a word, etc ([Benor and Levy, 2006](#)). However, that same study also showed that those constraints don’t account for the ordering preferences of high-frequency binomials like *salt and pepper*. For those binomials, the preferences can’t be predicted solely from the properties of the individual words, suggesting that the binomial is instead stored *holistically*. Binomial ordering preferences therefore show sensitivity to both abstract preferences and holistic storage depending on their frequency, which makes them a particularly useful case for studying the trade-off between the two.

1.2 Abstract Preferences in Language Models

Language models offer a way to study this trade-off that behavioral work cannot. When an expression stops being formed compositionally and starts being stored holistically is fundamentally a question about experience, but a human speaker’s history with any particular expression can only be estimated reasonably using corpus frequency. In a language model, that history is the training corpus itself, and it can be counted exactly. The representations are also directly inspectable, and the computation that produces them can be intervened on. Neither is possible for human learners.

The evidence for abstract preferences and holistic storage in language models has been more mixed than in the human behavioral literature. Models clearly generalize beyond what they have seen: [Misra and Mahowald \(2024\)](#) show that language models can produce the rare *a beautiful five days* construction even when it is withheld from training, generalizing to it from related constructions that are better attested, and [Yao et al. \(2025\)](#) find that dative alternation preferences are shaped by abstract constraints. At the same time, models retain a great deal item-specificity. [McCoy et al. \(2023\)](#) find that while model-generated text is of-

ten as structurally novel as human text, the same models also reproduce passages of over a thousand words verbatim from their training data. As such, it is clear that both abstract preferences and holistic storage are clearly at play within language models, however what remains unclear is how these types of knowledge trade off and when they emerge during training.

Perhaps most related to the present study, language models’ ordering preferences for binomials appear not to be driven by abstract preferences at all]. [Houghton et al. \(2025\)](#) examined whether language models showed the same abstract preferences as humans, and found that the models’ ordering preferences were driven exclusively by frequency statistics, with no evidence of an effect of abstract preferences. However, that finding faces limitations along two main dimensions. First, the study can only claim to find no evidence that the models’ ordering preferences were driven by *human* ordering preferences; it may very well be that their preferences are driven by abstract preferences that simply differ from those humans show. Second, the examination was conducted on model behavior, in the form of output probabilities, rather than over the representations that produce it. A model could in principle derive an ordering from abstract properties of the binomial and still produce output statistics that look frequency-driven.

1.3 Present Study

The present study thus asks whether binomial ordering preferences in language models are driven by abstract structure or if they are simply holistically stored. We make two changes to the design of [Houghton et al. \(2025\)](#). First, rather than testing model behavior against a fixed inventory of human constraints, we ask whether *any* systematic structure is recoverable at all, by training a probe to predict a model’s ordering preference from its hidden representations and evaluating it on binomials that the probe was never trained on. Second, we separate the two components architecturally rather than statistically. Zeroing out cross-word attention forces the model’s preference to be purely compositional: a strictly additive function of the individual words evaluated separately. Interventions on attention have been used to trace factual recall and to isolate circuits ([Geva et al., 2023](#); [Wang et al., 2023](#)), but to our knowledge this is the first use of one to separate holistic storage from abstract preferences. We apply this to 16 models across five

families, spanning human-scale to trillion-token training corpora, and to training checkpoints for two families matched in parameters but not in corpus size.

Our main contributions are:

- **Empirical evidence for both abstract preferences and frequency-driven holistic storage.** We show that a probe trained on LLM hidden-state representations for binomials can predict the ordering preferences for binomials it hasn't seen, however the binomial's relative frequency in the training data predicts a model's preference over and above anything those abstract preferences predict, and does so more strongly the more often the model encountered it.
- **An architectural intervention for isolating abstract preferences.** Zeroing out cross-word attention forces the representation into a strictly additive function of the two words, so the probe can only use what each word carries on its own.
- **Evidence that storage shifts lexical, not only phrasal, representations.** Zeroing out cross-word attention mitigates but does not eliminate the item-specific knowledge, indicating that repeated exposure to a phrase also changes the individual lexical representations themselves.

2 Methods

All three experiments apply the same procedure and differ only in which models¹ and checkpoints it is applied to. This section describes that procedure: the datasets, the probes, and the analyses.

2.1 Datasets

In order to examine the ordering preferences for binomials, we created two binomial corpora: a novel set and an attested set. The **novel set** comprises approximately 340,000 binomials from the Wikitext corpus (Merity et al., 2016) selected to be absent from BabyLM's training corpus. The **attested corpus set** comprises approximately 49,000 binomials from the BabyLM corpus (Warstadt et al., 2023).

The novel set is genuinely novel only for the BabyLM models; for the large-scale models it consists of infrequent attested binomials rather than

¹Appendix A lists all sixteen models with their parameter counts and training-token estimates.

unseen ones. We retain the label *novel* throughout, and return to what it licenses in the Limitations.

Both sets were extracted automatically, with a parser-based criterion requiring the two words to form a minimal coordinate phrase (Appendix B).

For each corpus binomial and each model, we also calculate the ordering preference, which is operationalized as the model's own log-probability ratio: $y_{\text{true}} = \log P(W_1 \text{ and } W_2) - \log P(W_2 \text{ and } W_1)$.

For the developmental analysis in Experiment 3, we probe six checkpoints each from the BabyLM and Pythia models. BabyLM checkpoints were sampled at approximately even intervals across training; Pythia checkpoints were instead chosen so that each model had seen roughly the same number of tokens as the corresponding BabyLM. These span roughly 20M to 2B effective training tokens.

2.2 Representation Extraction

For each model we extract final-layer hidden states, presenting each binomial in both orderings (W_1 and W_2 ; W_2 and W_1) within the same sentence frame. Each binomial is represented in two ways:

- **Binomial-level:** $[\bar{h}(W_1 \text{ and } W_2); \bar{h}(W_2 \text{ and } W_1)]$, where $\bar{h}$ denotes the mean of token representations across all tokens in the binomial in each ordering.
- **Word-level:** $[h(W_1); h(W_2)]$, where $h(W_k)$ is the hidden state of word W_k taken from the sentence in which it appears in first position.

2.3 Attention Conditions

We additionally probe representations under two attention conditions:

- **Full context:** standard forward pass with full cross-token attention. Each word's representation reflects contextual influences from all other (previous) tokens in the sentence.
- **Attention-zeroed:** the same forward pass, except that we zero out attention from the second word and the conjunction to the first word, at every layer (Appendix C).

This gives three representation types per model: binomial-level under each attention condition, and full-context word-level.

Because y_{true} is a difference of log-probabilities over the two orderings, and because causal masking already prevents W_1 from attending to W_2 ,

blocking attention from W_2 and the conjunction to W_1 leaves the model blind to the other word. The resulting preference is necessarily of the form $g(W_1) - g(W_2)$: an additive function of the two words evaluated separately, with no interaction term. The attention-zeroed model therefore reports not a degraded preference but the preference it would hold given only per-word information.

2.4 Probes

For each representation type, we train a two-layer MLP classifier (hidden dimension 128, ReLU activation) to predict binomial ordering preferences from the extracted representations. The probe takes the concatenated representations for a binomial and outputs a single continuous value $\hat{y}$, trained under mean-squared-error loss against y_{true} , with antisymmetric augmentation so that a swapped ordering carries the negated label. Optimizer settings, early stopping, and the augmentation rate are given in Appendix C. We report cross-validated R^2 (the squared Pearson correlation between $\hat{y}$ and y_{true} on held-out folds), averaged across 10 folds.²

We evaluate generalization under two 10-fold cross-validation conditions:

- **Binomial-held-out:** binomials are held out as units; each binomial in the test set is excluded from the training set, however overlapping words are allowed (e.g., *radios and computers* is not excluded from the training even if *radios and televisions* is in the test set).
- **Word-held-out:** binomials with overlapping words are excluded from the training set. For example, if *radios and televisions* is in the test set, then *radios and computers* is excluded from the training set.

Both splits are applied to the novel set; the attested set is not cross-validated, and enters only as the transfer target described below. We report fold standard deviations throughout, since word-held-out yields far fewer test items per fold (Appendix E).

Finally, following Hewitt and Liang (2019), we train a matched control probe on randomly permuted labels; near-zero control R^2 indicates the probe is recovering linguistic structure rather than label artifacts or probe capacity.

²The results below do not depend on the MLP’s capacity. A ridge regression on the same antisymmetric features, using the same folds and a single tuned hyperparameter, recovers most of the MLP’s performance. Full comparison in Appendix D.

2.5 Analyses

We report two measurements. The first is the probe’s cross-validated R^2 under the two splits above, of which word-held-out is the stricter (since binomial-held-out R^2 may be inflated by general lexical preferences).

The second measurement evaluates the probe on the attested set. Since that probe has seen only binomials the model encountered rarely or not at all, its prediction $\hat{y}$ for an attested binomial estimates what abstract knowledge alone would predict, and cannot reflect the probe (or the LLM) having memorized that binomial. In order to analyze the extent to which the model’s ordering preference within the attested set is driven by abstract vs holistic knowledge, we implement a Bayesian regression model using *brms* (Bürkner, 2017) with default priors. Our dependent variable is y_{true} , the model’s ordering preference. The probe’s prediction, $\hat{y}$, stands in for abstract preferences. Relative frequency, r , is the centered proportion of occurrences in alphabetical ordering: 0.5 means the binomial always occurs in alphabetical ordering in the corpus, 0 means it occurs evenly in both orderings, and -0.5 means it always occurs in non-alphabetical ordering. Letting f be log overall frequency (overall occurrences in the model’s corpus, logged³), our regression equation is:

$$\begin{aligned} y_{\text{true}} &\sim \beta_0 + \beta_1 \hat{y} + \beta_2 r + \beta_3 f \\ &\quad + \beta_4 (\hat{y} \times f) + \beta_5 (r \times f) \\ \log \sigma &= \gamma_0 + \gamma_1 f \end{aligned} \quad (1)$$

y_{true} and $\hat{y}$ are centered and scaled within each fit; f is centered but not scaled; r is left on its original scale, which already runs from -0.5 to $+0.5$ (Appendix F).

The residual scale is modeled rather than assumed constant. The spread of y_{true} varies with frequency, and an unmodeled trend there would inflate the slopes at one end of the range.

³Frequency is each binomial’s exposure in the model’s own training corpus: exact counts for BabyLM (Warstadt et al., 2023) and Pythia, and infini-gram Pile counts (Gao et al., 2020; Liu et al., 2024) standing in as a proxy for the rest. Appendix G gives the distributions.

3 Experiment 1: BabyLM Representations

In this first experiment, we probe three BabyLMs (125M, 350M, and 1.3B parameters)⁴ at their final training checkpoint. Because these models were trained on an amount of data comparable to humans, and because the novel set excludes binomials in their training corpus, they give the cleanest test of whether a language model encodes abstract binomial ordering preferences.

3.1 Results

3.1.1 Probe Performance

Results are shown in Figure 1; full numerical results are in Appendix H (Table 5). The control probe R^2 values were consistently near zero across all conditions and all three models (reported alongside each R^2 in Table 5), confirming that the probe doesn’t recover spurious signal from label noise.

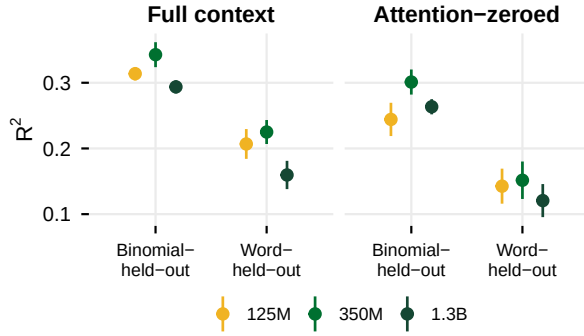


Figure 1: Cross-validated probe R^2 at the final hidden layer for the BabyLM models, binomial level, by attention condition. Bars = ± 1 SD across 10 folds.

For the binomial-level conditions, binomial-held-out R^2 ranges from 0.24 to 0.34; word-held-out R^2 is consistently lower, ranging from 0.12 to 0.23. The drop reflects a lexical generalization cost: some of the preference signal is tied to word identity rather than being fully abstract. Probe accuracy is non-monotonic with model size, and the full-context condition yields higher binomial-held-out R^2 than attention-zeroed, a difference we report but do not build on.

3.1.2 Abstract and Stored Contributions

Results are shown in Figure 2. Both abstract preferences and relative corpus frequency predict the

⁴The BabyLM models used here are OPT-based models trained for 20 epochs on the BabyLM v3 corpus (Houghton et al., 2026; Warstadt et al., 2023).

model’s ordering preference, and both do so credibly in all three models (Appendix F). The effect of relative frequency is stronger for high-frequency binomials, while the effect of abstract preferences is constant across different frequencies.

Zeroing out attention reduces both the effects of abstract preferences and relative frequency for the BabyLMs and additionally results in a negative interaction effect between abstract preferences and overall frequency, suggesting that abstract preferences are a *worse* predictor of ordering preferences for high-frequency binomials when attention is zeroed.

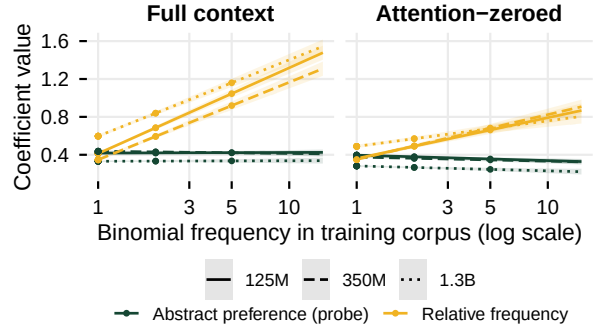


Figure 2: How strongly each source contributes to the BabyLM models’ ordering preference, by binomial frequency. Values are slopes from the fitted posteriors of Equation 1. Bands are 95% credible intervals.

4 Experiment 2: Cross-Family Final-Layer Representations

Experiment 1 demonstrates that binomial ordering preferences within the BabyLMs are driven by both abstract preferences and holistic storage. Those models are small in both parameters and training data, so Experiment 2 applies the same probe to 13 further models across four large-scale families: Pythia, GPT-2, OLMo, and Llama, with the BabyLM results carried over for comparison (Appendix A).

4.1 Results

4.1.1 Probe Performance

Figure 3 shows both cross-validation splits for all 16 models; full numbers, control probes, and the word-level replication are in Appendix I. Control R^2 was near zero throughout. All large-scale models substantially outperform BabyLM: full-context binomial-held-out R^2 spans 0.43 to 0.65 against 0.29–0.34, and word-held-out R^2 is lower than binomial-held-out for every model and condition.

Attention-zeroing raises binomial-held-out R^2 in every large-scale family, most sharply in GPT-2, where it reaches 0.79 at 117M against 0.65 under full context; Llama-3-8B is the sole exception. Scale effects within families are non-monotonic as a function of number of parameters.

4.1.2 Abstract and Stored Contributions

Results are shown in Figure 4. Similar to the BabyLMs, there is an effect of both abstract preferences and relative frequency for all models. Further, the effect of relative frequency is stronger for frequent binomials compared to infrequent ones. Unlike the BabyLMs, however, the effect of abstract preferences *decreases* as a function of frequency. In other words, the more the models encounter a binomial, the more their ordering preferences are governed by that binomial’s frequency and the less by abstract preferences.

Zeroing out attention mitigates the effect of relative frequency substantially for both the main-effect and the interaction with overall frequency, however both remain credible. Further, the effect of abstract preferences *increases* in every one of the large-scale models, while the interaction effect grows more negative in all but one. In other words, the effect of abstract preferences is generally stronger with attention zeroed out, but this increase is weaker for high-frequency binomials.

Notably, the effect of zeroing out attention differs between the BabyLMs and the other models. For the BabyLMs, zeroing out attention weakens the effect of both abstract preferences and relative frequency; for the large-scale models, it weakens the effect of relative frequency but *strengthens* the effect of abstract preferences. Further, the reduction in the effect of relative frequency is smaller for the BabyLMs than for the other models. In other words, abstract preferences appear to depend on cross-word attention for the BabyLMs, while for the large-scale models they are recoverable from the individual word representations alone.

5 Experiment 3: Developmental Trajectories

Experiments 1 and 2 establish that language models encode binomial preferences at the end of training, but leave open when those representations emerge, and whether the trade-off between abstract preferences and holistic storage is present early or accumulates with exposure.

The probe, input modes, attention conditions, and cross-validation splits are as described previously. We refit Equation 1 at each saved checkpoint of the three BabyLM models and the three Pythia models, in both attention conditions: 72 fits in all.

5.1 Results

Probe trajectories are shown in Figure 5 and the two sources’ contributions in Figure 6. BabyLM binomial-held-out R^2 is already high at the earliest checkpoint (~20M tokens: 0.67–0.72) and broadly declines to 0.32–0.38 by the end of training. At the earliest checkpoint, the attention-zeroed condition outperforms full context (0.78–0.80 vs. 0.67–0.72); this ordering reverses between ~120M and ~750M tokens, so that full context exceeds attention-zeroed at the final checkpoint for all three model sizes. Pythia shows an inverted-U trajectory instead: R^2 rises from 0.68–0.74 at 33.6M tokens to a peak of 0.80–0.86 at 537M tokens, then declines to 0.71–0.76 at 1074M–2097M tokens. The attention condition flip for Pythia follows the same qualitative pattern as BabyLM. Full numerical and word-level results are in Appendix J.

The effect of relative frequency is negligible across the first three checkpoints of every model. From the fourth checkpoint it is credibly positive in 36 of 36 model-by-condition cells and climbs thereafter: to 0.463, 0.414 and 0.802 in the three BabyLM models and to 0.488, 0.488 and 0.530 in the three Pythia models. This holds in both attention conditions, so what grows is not solely the binomial-specific representation.

At the fourth checkpoint, the first at which relative frequency is credibly positive in every model, attention-zeroing removes none of the effect. In other words, at the point where storage is first measurable, it survives an intervention that removes the phrasal representation, so what has shifted at that point is the lexical representations. Storage in the phrasal representation then accumulates on top of it. The share of the relative-frequency effect that attention-zeroing removes rises from nothing at the fourth checkpoint to 20–23% by the last in the three Pythia models and 8–24% in the three BabyLM models, and reaches 40–70% in the fully trained large-scale models. The interaction effect between relative frequency and overall frequency follows the same course, reaching 0.142, 0.103 and 0.291 in BabyLM and 0.052, 0.042 and 0.054 in Pythia.

The effect of abstract preferences moves inde-

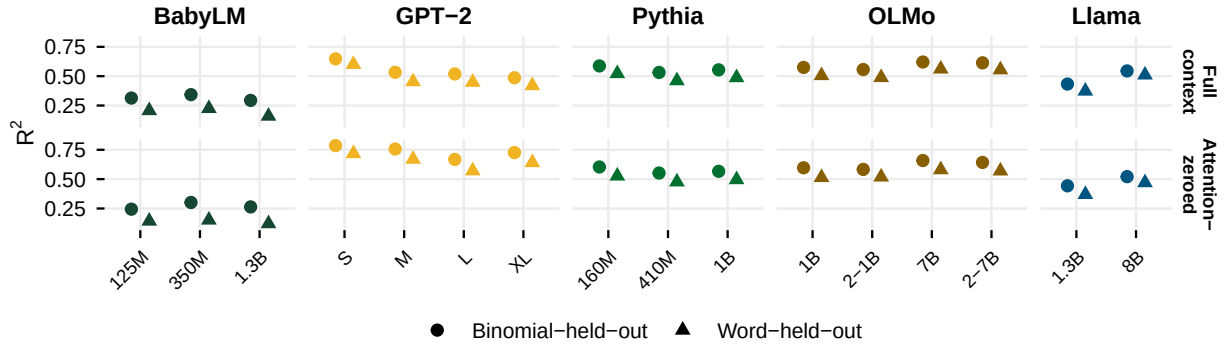


Figure 3: Cross-validated R^2 for all 16 models, by attention condition and cross-validation split. Circles = binomial-held-out; triangles = word-held-out. Bars = 95% CI across 10 folds. Models ordered by parameter count within each family and colored by family.

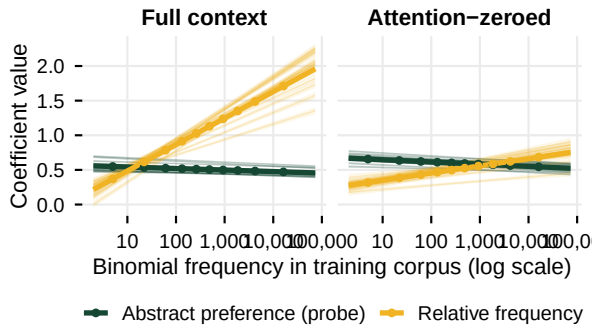


Figure 4: How strongly each source contributes to the ordering preference, by binomial frequency, for the 13 large-scale models. Values are simple slopes from the fitted posteriors of Equation 1. Faint lines are individual models with their 95% credible intervals, bold lines their mean.

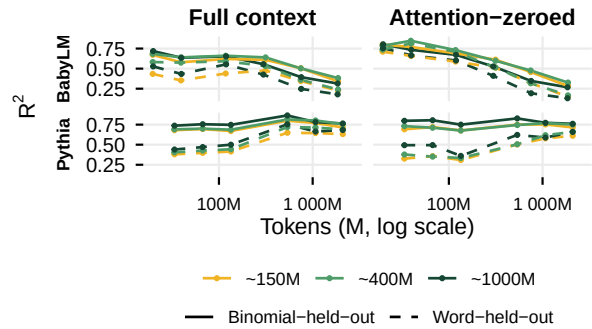


Figure 5: Cross-validated probe R^2 across BabyLM and Pythia training checkpoints, by attention condition and cross-validation split. Control $R^2 \approx 0$.

pendently of all this. It declines steadily across BabyLM training in both conditions. For Pythia it rises over the early window, then falls below its starting value by the final model. Its interaction with overall frequency is credibly negative under attention-zeroing in 26 of the 36 checkpoint cells.

6 Discussion and Conclusion

The present study examined whether language models’ binomial ordering preferences are driven by abstract structure, and if so, how that structure emerges throughout training. Across 16 models from five families, we found evidence for both abstract and item-specific sources of ordering preference. Preferences generalize to binomials composed entirely of words absent from training, demonstrating abstract structure that is shared across independently trained models. At the same time, a binomial’s relative frequency contributes over and above abstract preferences, and contributes more as the binomial becomes more

frequent. These two sources also behave differently over training: abstract preferences are present early, whereas the item-specific contribution only becomes measurable later, which bounds when it can be observed rather than when it arises. Finally, zeroing out cross-word attention removes only part of the relative-frequency contribution, suggesting that storage shifts the lexical representations as well as the phrasal one.

The evidence for abstract preferences is particularly strong because it does not depend on generalization across familiar words alone. The probe recovers a model’s preference for binomials built entirely from words it was never trained on. A ridge probe with a single tuned hyperparameter recovers the same pattern, showing that the result does not depend on the MLP’s capacity (Appendix D). The answer to the question left open by Houghton et al. (2025) is therefore that models do have abstract preferences, but ones that a test for human constraints failed to detect. Further evidence that these preferences are systematic comes without the probe at all: independently trained models converge on the same preferences for the same novel

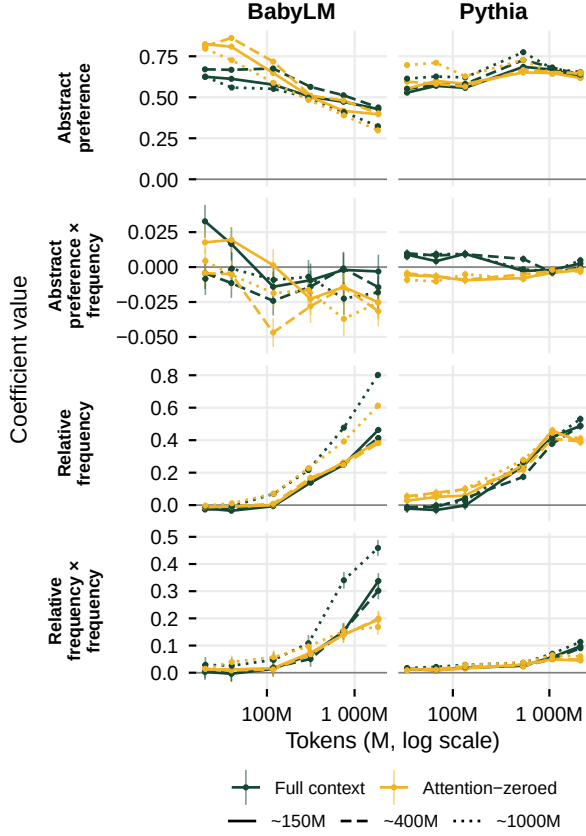


Figure 6: All four terms of Equation 1 across training, with 95% credible intervals. Rows are the two sources and their interactions with overall frequency; color marks the attention condition and linetype the model size.

binomials ($r = 0.45$, against a shuffled baseline of $r \approx 0.00$). This agreement holds even among the three BabyLM models, which share a training corpus but differ in capacity, and is stronger within model families than between families. Because all sixteen models received the same sentence frame, the latter pattern cannot be attributed to the frame itself.

Abstract structure, however, does not fully determine a model’s preference for a particular binomial. A binomial’s relative frequency predicts the model’s preference over and above what the abstract structure predicts, and this item-specific contribution increases with frequency. Unlike the abstract structure, this information is specific to the particular binomial. The result therefore reveals a frequency-dependent division between generalizable structure and experience with individual items, closely paralleling the trade-off between abstract preferences and item-specific experience observed by Morgan and Levy (2016) in humans.

Abstract preferences are present from the be-

ginning of training. Probe R^2 is already high at the earliest BabyLM checkpoint, at roughly 20M tokens, and the effect of abstract preferences in Equation 1 is large from the first checkpoint in every model. In the BabyLMs it then declines as the item-specific effect grows, potentially due to the lack of diversity in the training data adding item-specific knowledge without containing additional information about abstract preferences. The item-specific contribution follows a different trajectory: it is negligible through the first three checkpoints of every model, becomes credibly positive everywhere from the fourth, and then climbs monotonically under full context without leveling off. This is not conclusive evidence against item-specific knowledge being learned early, however, since the measurement relies on a frequency-difference in representations to detect holistic storage. The absence of a frequency difference, consequently, does not necessarily indicate lack of storage.

Finally, the attention intervention provides further insight into how this item-specific information is represented. Zeroing out cross-word attention mitigates but does not eliminate the contribution of relative frequency: at the first checkpoint where a binomial’s relative frequency predicts the model’s ordering preference at all, attention-zeroing removes none of that prediction; the share removed then increases over training. This pattern suggests that item-specific information is not exclusively stored as a holistic representation of the binomial. Instead, it seems possible that repeated exposure to a binomial gradually shifts the representations of its individual words, so that a word encountered overwhelmingly in first position comes to encode something of that tendency on its own. Such information remains item-specific in that it concerns particular words and their learned positional tendencies rather than a general constraint, but it no longer requires the model to represent the specific binomial context. On this interpretation, entrenchment shifts the lexical representations before it shifts the phrasal one. It isn’t surprising that the phrasal representation comes to carry ordering information, since attention is what builds a representation of the pair in the first place. The more interesting result is that the lexical representations shift too. This parallels evidence from human processing that experience with a phrase alters the representations of its individual words (Houghton, 2025b).

Overall, the findings bridge a longstanding gap between artificial language models and usage-

based accounts of language. Such accounts have long proposed that frequent expressions become increasingly entrenched, with behavioral work on binomials showing that human ordering preferences reflect both abstract constraints and holistic storage. The present results show a similar division emerging in a system with no innate linguistic preferences: trained only to predict text, the models develop both generalizable ordering preferences and frequency-sensitive knowledge of individual expressions. The results therefore suggest that the interaction between abstraction and entrenchment need not be explicitly built into a learner. It can emerge from exposure itself, with generalizable structure and item-specific knowledge developing together but on different timescales and, potentially, in different parts of the representation.

Limitations

The novel binomials were selected to be absent from BabyLM’s training corpus, so they are genuinely novel only for the three BabyLM models; indexed against Dolma (Soldaini et al., 2024), a trillion-token corpus, roughly 98.4% occur at non-zero frequency. For the thirteen large-scale models, then, the probe is trained on infrequent attested binomials, not unseen ones. This is why Experiment 1 anchors the interpretation. If a recovered preference must have been generalized and not retrieved, that inference rests on the BabyLMs; the larger models extend the pattern but don’t establish it. Corpus frequency is likewise an exact measure of what the model saw only for BabyLM and Pythia; for GPT-2, OLMo, and Llama, Pile counts stand in as a proxy for web-scale exposure.

Additionally, the results are limited to English and to a single construction, and checkpoints were only examined for two of the five families, so the developmental claims rest on BabyLM and Pythia. Whether attention-zeroing separates per-word from relational information as cleanly in constructions with a different structure is untested, though the argument that it should is architectural rather than empirical and does not depend on properties peculiar to binomials.

Finally, the intervention separates additive from non-additive information, not abstract from holistic information directly. The two line up only if abstract constraints can be stated over each word on its own. Most of the constraints in Morgan and Levy (2016) can be: relative word length or fre-

quency compares two quantities that each word carries by itself, so an additive representation preserves them. A constraint over the juncture between the two words is different. Avoiding a stress clash across the boundary, for instance, is abstract and generalizable, but the mask would remove it and we would count it as holistic storage. The dissociation between the two effects suggests this isn’t what’s driving our results, since losing information indiscriminately would have weakened both effects rather than only one. Even so, the design doesn’t rule it out.

References

- Sarah Bunin Benor and Roger Levy. 2006. [The chicken or the egg? a probabilistic analysis of english binomials](#). *Language*, 82(2):233–278.
- Jean Berko. 1958. [The child’s learning of english morphology](#). *WORD*, 14(2-3):150–177.
- J. Kathryn Bock. 1986. [Syntactic persistence in language production](#). *Cognitive Psychology*, 18(3):355–387.
- Joan Bybee and Joanne Scheibman. 1999. [The effect of usage on degrees of constituency: the reduction of don’t in english](#). *Linguistics*, 37(4).
- Paul-Christian Bürkner. 2017. [brms: An R package for Bayesian multilevel models using Stan](#). *Journal of Statistical Software*, 80(1):1–28.
- Leo Gao, Stella Biderman, Sid Black, Laurence Golding, Travis Hoppe, Charles Foster, Jason Phang, Horace He, Anish Thite, Noa Nabeshima, Shawn Presser, and Connor Leahy. 2020. [The pile: An 800GB dataset of diverse text for language modeling](#). *Preprint*, arXiv:2101.00027.
- Mor Geva, Jasmijn Bastings, Katja Filippova, and Amir Globerson. 2023. [Dissecting recall of factual associations in auto-regressive language models](#). In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 12216–12235, Singapore. Association for Computational Linguistics.
- John Hewitt and Percy Liang. 2019. [Designing and interpreting probes with control tasks](#). In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, pages 2733–2743, Hong Kong, China. Association for Computational Linguistics.
- Matthew Honnibal, Ines Montani, Sofie Van Landeghem, and Adriane Boyd. 2020. [spaCy: Industrial-strength natural language processing in python](#). Zenodo.

745	Zachary Nicholas Houghton. 2025a. The bolts and nuts of language processing: An investigation into the noisy-channel processing of binomials . Master's thesis, University of California, Davis.	801
746		802
747		803
748		804
749	Zachary Nicholas Houghton. 2025b. Multi-Word Representations in Minds and Models: Investigating the Storage of Multi-Word Phrases in Humans and Large Language Models . Ph.D. thesis, University of California, Davis.	805
750		806
751		807
752		808
753		809
754	Zachary Nicholas Houghton and Vsevolod Kapatsinski. 2026. Exemplars in disguise: Pure exemplar models mimic abstraction-first learning . <i>Preprint</i> , arXiv:2608.00821.	810
755		811
756		812
757		813
758	Zachary Nicholas Houghton and Emily Morgan. 2023. Does predictability drive the holistic storage of compound nouns? In <i>Proceedings of the Annual Meeting of the Cognitive Science Society</i> , volume 45.	814
759		815
760		816
761		817
762	Zachary Nicholas Houghton and Emily Morgan. 2024. Frequency-dependent preference extremity arises from a noisy-channel processing model . In <i>Proceedings of the Annual Meeting of the Cognitive Science Society</i> , volume 46.	818
763		819
764		820
765		821
766		822
767	Zachary Nicholas Houghton, Kenji Sagae, and Emily Morgan. 2025. The role of abstract representations and observed preferences in the ordering of binomials in large language models. In <i>Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 2: Short Papers)</i> , pages 695–702.	823
768		824
769		825
770		826
771		827
772		828
773		829
774	Zachary Nicholas Houghton, Yu Zhou, Dan Pluth, Jordan Hosier, and Vijay K. Gurbani. 2026. The holistic storage of verb+up phrases in text-based and audio-based language models . <i>Preprint</i> , arXiv:2606.13993.	830
775		831
776		832
777		833
778	Vsevolod Kapatsinski and Joshua Radicke. 2009. Frequency and the emergence of prefabs: Evidence from monitoring . In <i>Formulaic Language, Volume 2: Acquisition, Loss, Psychological Reality, and Functional Explanations</i> , volume 83 of <i>Typological Studies in Language</i> , pages 499–522. John Benjamins Publishing Company.	834
779		835
780		836
781		837
782		838
783		839
784		
785	Nikita Kitaev, Steven Cao, and Dan Klein. 2019. Multi-lingual constituency parsing with self-attention and pre-training . In <i>Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics</i> , pages 3499–3505, Florence, Italy. Association for Computational Linguistics.	840
786		841
787		842
788		843
789		844
790		845
791	Nikita Kitaev and Dan Klein. 2018. Constituency parsing with a self-attentive encoder . In <i>Proceedings of the 56th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)</i> , pages 2676–2686, Melbourne, Australia. Association for Computational Linguistics.	846
792		847
793		848
794		849
795		
796		850
797	Jiacheng Liu, Sewon Min, Luke Zettlemoyer, Yejin Choi, and Hannaneh Hajishirzi. 2024. Infini-gram: Scaling unbounded n-gram language models to a trillion tokens . <i>CoRR</i> , abs/2401.17377.	851
798		852
799		853
800		854
		855
		856
	R. Thomas McCoy, Paul Smolensky, Tal Linzen, Jianfeng Gao, and Asli Celikyilmaz. 2023. How much do language models copy from their training data? evaluating linguistic novelty in text generation using raven . <i>Transactions of the Association for Computational Linguistics</i> , 11:652–670.	
	Stephen Merity, Caiming Xiong, James Bradbury, and Richard Socher. 2016. Pointer sentinel mixture models . <i>Preprint</i> , arXiv:1609.07843.	
	David E. Meyer and Roger W. Schvaneveldt. 1971. Facilitation in recognizing pairs of words: evidence of a dependence between retrieval operations . <i>Journal of experimental psychology</i> , 90(2):227.	
	Kanishka Misra and Kyle Mahowald. 2024. Language models learn rare phenomena from less rare phenomena: The case of the missing AANNs . In <i>Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing</i> , pages 913–929, Miami, Florida, USA. Association for Computational Linguistics.	
	Emily Morgan and Roger Levy. 2015. Modeling idiosyncratic preferences: How generative knowledge and expression frequency jointly determine language structure. In <i>Proceedings of the Annual Meeting of the Cognitive Science Society</i> , volume 37.	
	Emily Morgan and Roger Levy. 2016. Abstract knowledge versus direct experience in processing of binomial expressions . <i>Cognition</i> , 157:384–402. PMID: 27776281 Publisher: The Authors.	
	Emily Morgan and Roger Levy. 2024. Productive knowledge and item-specific knowledge trade off as a function of frequency in multiword expression processing . <i>Language</i> , 100(4):e195–e224.	
	Luca Soldaini, Rodney Kinney, Akshita Bhagia, Dustin Schwenk, David Atkinson, Russell Authur, Ben Bogin, Khyathi Chandu, Jennifer Dumas, Yanai Elazar, and 1 others. 2024. Dolma: An open corpus of three trillion tokens for language model pretraining research . <i>arXiv preprint arXiv:2402.00159</i> .	
	Joseph P. Stemberger and Brian MacWhinney. 2004. Are inflected forms stored in the lexicon . <i>Morphology: Critical concepts in linguistics</i> , 6:107–122.	
	Kevin Ro Wang, Alexandre Variengien, Arthur Conmy, Buck Shlegeris, and Jacob Steinhardt. 2023. Interpretability in the wild: a circuit for indirect object identification in GPT-2 small . In <i>The Eleventh International Conference on Learning Representations, ICLR 2023, Kigali, Rwanda, May 1-5, 2023</i> . OpenReview.net.	
	Alex Warstadt, Aaron Mueller, Leshem Choshen, Ethan Wilcox, Chengxu Zhuang, Juan Ciro, Rafael Mosquera, Bhargavi Paranjabe, Adina Williams, Tal Linzen, and Ryan Cotterell. 2023. Findings of the BabyLM challenge: Sample-efficient pretraining on developmentally plausible corpora . In <i>Proceedings of the BabyLM Challenge at the 27th Conference on</i>	

Computational Natural Language Learning, pages
1–34, Singapore. Association for Computational Lin-
guistics.

Qing Yao, Kanishka Misra, Leonie Weissweiler, and
Kyle Mahowald. 2025. [Both direct and indirect evi-
dence contribute to dative alternation preferences in
language models](#). *CoRR*, abs/2503.20850.

A Model Details

Table 1 lists all 16 models used across experiments, organized by family, with approximate parameter counts and training token counts.

Table 1: Models used across experiments. Training token counts for full-scale models are approximate; GPT-2 counts are estimated from the reported 40GB WebText corpus. BabyLM models train for 20 epochs over 150M tokens (3B effective token exposures).

Family	Model	Parameters	Tokens
BabyLM	BabyLM-125M	125M	150M
	BabyLM-350M	350M	150M
	BabyLM-1.3B	1.3B	150M
GPT-2	GPT-2	117M	~8B
	GPT-2-medium	345M	~8B
	GPT-2-large	762M	~8B
	GPT-2-XL	1.5B	~8B
Pythia	Pythia-160M	160M	~300B
	Pythia-410M	410M	~300B
	Pythia-1B	1B	~300B
OLMo	OLMo-1B	1B	3T
	OLMo-7B	7B	2.5T
	OLMo-2-1B	1B	~4T
	OLMo-2-7B	7B	~4T
Llama	Llama-1.3B	1.2B	~9T
	Llama-3-8B	8B	~15T

B Binomial Extraction Criteria

Candidate binomials (“W1 and W2”) were first selected using spaCy (Honnibal et al., 2020) part-of-speech tags, retaining only cases where both W_1 and W_2 are open-class words (nouns, verbs, adjectives, or adverbs). Each candidate was then parsed with the Berkeley Neural Parser (Kitaev et al., 2019; Kitaev and Klein, 2018) to confirm that the two words form a minimal coordinate phrase. The inclusion criterion was that the two words be co-constituents under the same phrasal node (NP, VP, ADJP, ADVP, PP, or equivalent), with that node spanning exactly the three tokens [W_1 , and, W_2] and nothing else. This ensures the extracted items are genuine coordinate binomials rather than incidental sequences where “and” links clauses or longer phrases. Novel binomials were extracted from Wikitext by the identical procedure, with the added requirements that both words appear in BabyLM’s vocabulary and that the binomial is absent from the attested set.

Because our extraction criterion requires W1 and W2 to form a phrase-level constituent spanning exactly W1, “and,” and W2, binomials that do not satisfy this constituency criterion are excluded. We do not additionally require the two conjuncts to receive the same POS tag, because spaCy’s POS tagger can misclassify participial forms in coordinate contexts, for example tagging “mortified” as VERB rather than ADJ in “disgusted and mortified.” Requiring POS-tag agreement would therefore risk excluding valid binomials because of POS-tagging errors.

C Implementation Details

Probe training. The probe is optimized with Adam (learning rate 10^{-3} , weight decay 10^{-4}) under mean-squared-error loss, with early stopping (patience 20) against a 10% held-out validation split. Antisymmetric augmentation is applied with probability 0.5: for each binomial the flipped version, with swapped ordering and negated label, is added to the training batch. The input dimensionality follows from the representation type.

Attention-zeroing. The zero weight is imposed before the softmax, by adding a large negative constant to the corresponding attention logits, so that the remaining positions renormalize to sum to one. Setting the weights to zero after the softmax instead would leave them summing to less than one, which would scale down the attention output and make the intervention a partial ablation rather than a clean removal of one word. The mask is supplied as a 4D attention mask rather than a 2D padding mask, so it applies at every transformer layer rather than only the first. This matters because a mask applied at the first layer alone would leave W_2 able to recover information about W_1 indirectly at greater depth, through positions that did attend to it.

Recomputing the target. Because y_{true} is the model’s own log-probability ratio, it is recomputed from whichever forward pass produced the representations being probed. The two full-context representation types therefore share a target, while the attention-zeroed condition has its own: blocking cross-word attention changes the model’s output probabilities as well as its hidden states. Each attention condition is thus a self-consistent probe problem rather than two input encodings evaluated against a fixed target.

D Linear Probe Replication

The MLP probe has many free choices: hidden dimension, depth, activation, learning rate, weight decay, patience, validation fraction, augmentation probability, batch size, and initialization. A skeptical reading is that some of them were tuned toward the reported result. We therefore repeat the analysis with a probe that has one.

The linear probe is a ridge regression on the same features, fit on the same cross-validation folds. Its only free variable is the ridge penalty, selected per fold from $\{10^0, \dots, 10^5\}$ on an inner split of that fold's training data. That inner split is the same size as the MLP's early-stopping split, so both probes receive an identical budget of held-out data for hyperparameter selection.

Across 52 model-checkpoint cells and 207 comparisons, the two probes agree closely ($r = 0.928$ between their R^2 values). On the word-held-out split the linear probe recovers 86% of the MLP's R^2 on average (0.410 vs. 0.476).

Table 2: Linear (ridge) versus MLP probe R^2 at the final hidden layer, binomial-level input mode. Both probes use identical cross-validation folds. B = binomial-held-out; W = word-held-out. Gap = MLP minus linear.

Model	Condition	Split	Linear	MLP	Gap
BabyLM-125m	Attention-zeroed	B	0.169	0.244	+0.075
		W	0.103	0.143	+0.040
	Full context	B	0.197	0.314	+0.116
		W	0.147	0.207	+0.060
BabyLM-1_3b	Attention-zeroed	B	0.185	0.263	+0.078
		W	0.094	0.121	+0.027
	Full context	B	0.188	0.294	+0.105
		W	0.127	0.160	+0.032
BabyLM-350m	Attention-zeroed	B	0.211	0.301	+0.090
		W	0.121	0.152	+0.031
	Full context	B	0.223	0.343	+0.120
		W	0.170	0.225	+0.055
Llama-3.2-1B	Attention-zeroed	B	0.336	0.443	+0.107
		W	0.288	0.370	+0.082
	Full context	B	0.357	0.433	+0.076
		W	0.332	0.373	+0.041
Meta-Llama-3-8B	Attention-zeroed	B	0.465	0.521	+0.057
		W	0.420	0.470	+0.050
	Full context	B	0.507	0.545	+0.038
		W	0.485	0.510	+0.026
OLMo-1B-hf	Attention-zeroed	B	0.482	0.597	+0.114
		W	0.438	0.514	+0.076
	Full context	B	0.466	0.575	+0.108
		W	0.439	0.505	+0.066
OLMo-2-0425-1B	Attention-zeroed	B	0.477	0.583	+0.105
		W	0.448	0.518	+0.070
	Full context	B	0.460	0.557	+0.096
		W	0.438	0.488	+0.051
OLMo-2-1124-7B	Attention-zeroed	B	0.566	0.642	+0.076
		W	0.525	0.570	+0.044
	Full context	B	0.559	0.614	+0.055
		W	0.531	0.554	+0.023
OLMo-7B-hf	Attention-zeroed	B	0.575	0.658	+0.083
		W	0.518	0.580	+0.062
	Full context	B	0.553	0.621	+0.067
		W	0.518	0.561	+0.043
gpt2	Attention-zeroed	B	0.647	0.786	+0.139
		W	0.606	0.718	+0.111
	Full context	B	0.569	0.648	+0.079
		W	0.556	0.601	+0.045

(continued)

Model	Condition	Split	Linear	MLP	Gap
gpt2-large	Attention-zeroed	B	0.523	0.668	+0.145
		W	0.468	0.571	+0.103
	Full context	B	0.414	0.519	+0.105
		W	0.392	0.449	+0.057
gpt2-medium	Attention-zeroed	B	0.621	0.756	+0.136
		W	0.567	0.670	+0.103
	Full context	B	0.396	0.533	+0.138
		W	0.373	0.453	+0.081
gpt2-xl	Attention-zeroed	B	0.609	0.726	+0.117
		W	0.560	0.641	+0.081
	Full context	B	0.394	0.487	+0.092
		W	0.369	0.420	+0.051
pythia-160m	Attention-zeroed	B	0.481	0.604	+0.122
		W	0.456	0.527	+0.071
	Full context	B	0.459	0.587	+0.128
		W	0.445	0.522	+0.077
pythia-1b	Attention-zeroed	B	0.479	0.567	+0.088
		W	0.437	0.496	+0.058
	Full context	B	0.471	0.554	+0.083
		W	0.440	0.488	+0.048
pythia-410m	Attention-zeroed	B	0.434	0.552	+0.118
		W	0.407	0.477	+0.070
	Full context	B	0.413	0.532	+0.118
		W	0.392	0.461	+0.069

The gap between the probes is consistently larger on the binomial-held-out split than on the word-held-out split. This is expected: binomial-held-out performance is partly recoverable from word identity alone, which a flexible model can exploit and a linear map on the same features cannot. Word-held-out performance, on which the paper’s claims rest, is where the two agree most closely.

E Cross-Validation Split Sizes

Cross-validation is carried out on the novel set only. The attested set is never cross-validated: it is used solely as the transfer target described in Section 2.5. Both splits use ten folds, and the same folds are reused across models, attention conditions, and probe architectures so that all reported R^2 values are directly comparable.

Under the binomial-held-out split, binomials are assigned to folds directly, so each fold trains on nine tenths of the set and tests on the remaining tenth. Under the word-held-out split it is the *vocabulary* that is partitioned into ten folds. A binomial enters the test set for fold k only if both of its words are in fold k , and enters the training set only if neither word is. Binomials with one word in fold k and one word outside it are used in neither, and are discarded for that fold. Since a binomial needs both words drawn from the same tenth of the vocabulary to be testable, the word-held-out test set is roughly a hundredth of the set rather than a tenth. Table 3 gives the resulting sizes.

Table 3: Cross-validation fold sizes for the novel set (340,042 binomials over 50,137 unique words). Ranges are across the 10 folds. Discarded items under word-held-out are binomials with exactly one word in the held-out vocabulary fold.

Split	Train / fold	Test / fold	Discarded / fold
Binomial-held-out	306,038	34,004	0
Word-held-out	271,560–279,306	3,086–3,839	61,101 (18.0%)

F Frequency-Dependent Preference Regression

Full results for Equation 1: abstract preferences $\hat{y}$ (y_{pred}), the binomial’s relative frequency in the corpus (rel_freq), overall binomial frequency ($\log_freq$), the two moderations, and the residual-scale term.

The two kinds of variable are treated differently, and deliberately so. y_{true} and y_{pred} are properties of the model: both are log-probability ratios, so a unit is nominally the same everywhere, but the spread of that quantity reflects how sharp a given model’s output distribution is rather than how much it knows about binomials. Across the models here the standard deviation of y_{true} ranges from 2.71 to 4.85, and it runs opposite to probe R^2 , so a raw coefficient would partly measure output calibration. We therefore z-score both within each fit, which puts every coefficient in units of that model’s own preference spread.

rel_freq and $\log_freq$, by contrast, are properties of the training corpus: a proportion of occurrences and a count of them mean the same thing however they are measured, and dividing either by a corpus-specific standard deviation would replace a shared unit with a local one. They are therefore left on their own scales: rel_freq is a proportion running from -0.5 to $+0.5$ and $\log_freq$ is centered but not scaled, so both mean the same thing in every corpus. A rel_freq coefficient is therefore the change in ordering preference, in standard deviations, between a binomial always attested in one order and one always attested in the other; a $\log_freq$ coefficient is the change per unit of log frequency, which is a factor of $e \approx 2.72$ in the raw count; multiplying by 0.69 converts it to the change per doubling.

Because $\log_freq$ is centered within each fit, a main effect is the effect at that corpus’s average binomial. The two predictors are on different scales, so their coefficients should not be compared with each other.

Table 4: Frequency-dependent preference regression (Equation 1) for all 16 models at their final checkpoint, by attention condition. y_{true} and y_{pred} are z-scored within each fit; rel_freq is a proportion from -0.5 to $+0.5$ and $\log_freq$ is centered but not scaled, so both mean the same thing in every corpus. 95% credible intervals in brackets. Holistic storage predicts a negative $y_{\text{pred}} \times \log_freq$ and a positive $\text{rel_freq} \times \log_freq$, both attenuating under attention-zeroing.

Model	Condition	y_{pred}	rel_freq	$y_{\text{pred}} \times \log_freq$	$\text{rel_freq} \times \log_freq$
BabyLM-125M	Full context	0.418 [0.411, 0.426]	0.535 [0.519, 0.551]	0.003 [-0.010, 0.015]	0.393 [0.363, 0.423]
	Attention-zeroed	0.388 [0.380, 0.396]	0.419 [0.403, 0.436]	-0.025 [-0.036, -0.013]	0.187 [0.158, 0.217]
BabyLM-350M	Full context	0.434 [0.426, 0.442]	0.458 [0.442, 0.475]	-0.010 [-0.022, 0.002]	0.355 [0.325, 0.385]
	Attention-zeroed	0.370 [0.362, 0.378]	0.415 [0.398, 0.432]	-0.019 [-0.030, -0.007]	0.206 [0.176, 0.235]
BabyLM-1.3B	Full context	0.332 [0.325, 0.340]	0.706 [0.690, 0.722]	0.002 [-0.010, 0.015]	0.350 [0.320, 0.379]
	Attention-zeroed	0.275 [0.267, 0.283]	0.525 [0.509, 0.543]	-0.022 [-0.035, -0.010]	0.117 [0.087, 0.147]
GPT-2	Full context	0.489 [0.482, 0.497]	1.013 [0.988, 1.039]	-0.006 [-0.009, -0.003]	0.198 [0.189, 0.207]
	Attention-zeroed	0.683 [0.676, 0.689]	0.309 [0.286, 0.331]	-0.018 [-0.020, -0.015]	0.026 [0.018, 0.034]
GPT-2-M	Full context	0.447 [0.439, 0.454]	1.145 [1.119, 1.172]	-0.006 [-0.009, -0.003]	0.204 [0.195, 0.213]
	Attention-zeroed	0.640 [0.633, 0.646]	0.428 [0.404, 0.452]	-0.011 [-0.014, -0.009]	0.035 [0.027, 0.043]
GPT-2-L	Full context	0.454 [0.447, 0.462]	1.152 [1.125, 1.178]	-0.008 [-0.011, -0.005]	0.197 [0.187, 0.206]
	Attention-zeroed	0.566 [0.559, 0.574]	0.497 [0.472, 0.523]	-0.014 [-0.017, -0.011]	0.043 [0.034, 0.052]
GPT-2-XL	Full context	0.447 [0.439, 0.454]	1.191 [1.165, 1.218]	-0.009 [-0.012, -0.006]	0.197 [0.187, 0.206]
	Attention-zeroed	0.605 [0.598, 0.613]	0.458 [0.434, 0.482]	-0.014 [-0.017, -0.012]	0.048 [0.039, 0.057]
Pythia-160M	Full context	0.462 [0.454, 0.469]	1.060 [1.033, 1.086]	-0.003 [-0.006, -0.000]	0.183 [0.174, 0.192]
	Attention-zeroed	0.506 [0.499, 0.514]	0.557 [0.531, 0.583]	-0.008 [-0.011, -0.005]	0.033 [0.024, 0.042]
Pythia-410M	Full context	0.453 [0.445, 0.460]	1.172 [1.145, 1.198]	-0.011 [-0.013, -0.008]	0.188 [0.180, 0.197]
	Attention-zeroed	0.517 [0.510, 0.525]	0.579 [0.553, 0.605]	-0.016 [-0.019, -0.013]	0.053 [0.043, 0.062]
Pythia-1B	Full context	0.493 [0.486, 0.500]	1.127 [1.101, 1.153]	-0.011 [-0.014, -0.008]	0.166 [0.157, 0.174]
	Attention-zeroed	0.575 [0.568, 0.582]	0.523 [0.498, 0.548]	-0.014 [-0.017, -0.011]	0.047 [0.038, 0.055]
OLMo-1B	Full context	0.534 [0.527, 0.541]	1.052 [1.027, 1.077]	-0.011 [-0.014, -0.008]	0.161 [0.152, 0.169]
	Attention-zeroed	0.592 [0.585, 0.599]	0.551 [0.526, 0.577]	-0.012 [-0.015, -0.009]	0.049 [0.040, 0.058]
OLMo-7B	Full context	0.582 [0.575, 0.588]	0.989 [0.965, 1.012]	-0.009 [-0.011, -0.006]	0.137 [0.129, 0.146]
	Attention-zeroed	0.636 [0.630, 0.643]	0.558 [0.535, 0.581]	-0.011 [-0.014, -0.009]	0.055 [0.047, 0.063]
OLMo-2-1B	Full context	0.519 [0.512, 0.526]	1.058 [1.032, 1.082]	-0.012 [-0.015, -0.009]	0.140 [0.132, 0.149]
	Attention-zeroed	0.592 [0.585, 0.599]	0.544 [0.519, 0.568]	-0.015 [-0.018, -0.012]	0.043 [0.035, 0.052]

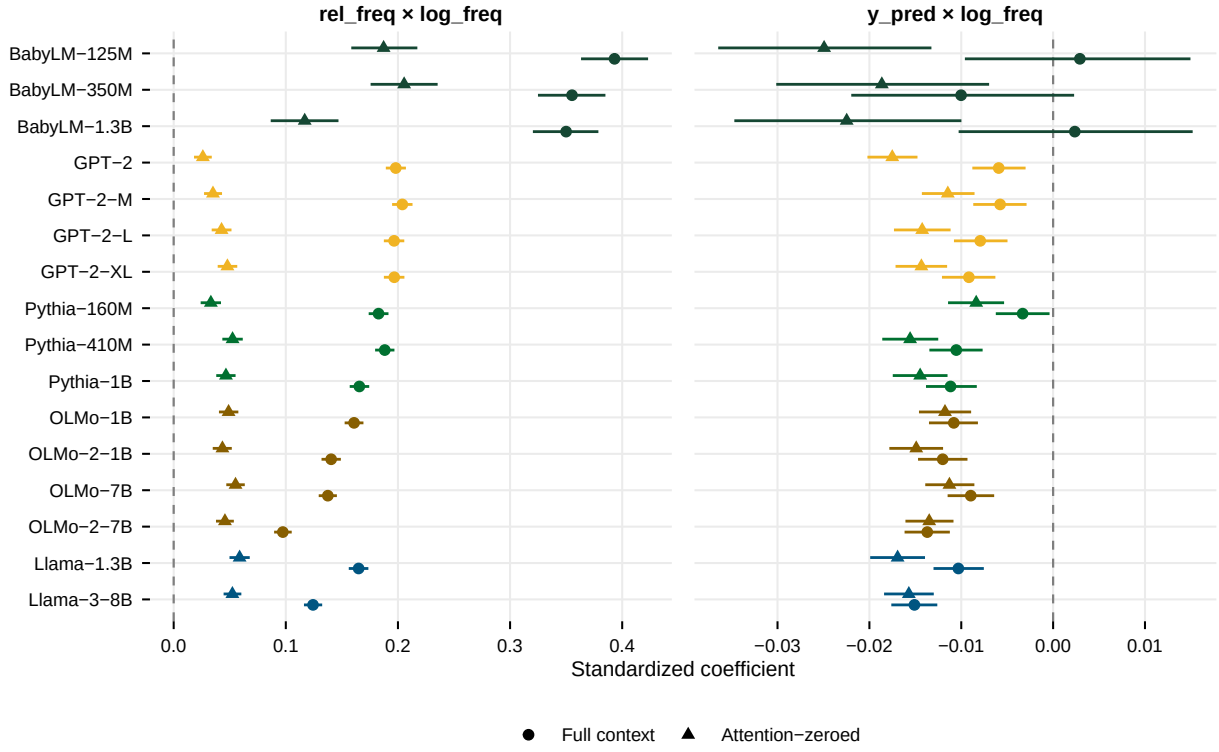


Figure 7: Moderation terms of Equation 1 by model and attention condition. Circles = full context; triangles = attention-zeroed. Points = posterior mean; bars = 95% credible intervals. Left: relative frequency matters more for frequent binomials, and attention-zeroing removes most of that. Right: abstract preferences matter less for frequent binomials, and attention-zeroing does not remove it.

(continued)

Model	Condition	y_pred	rel_freq	y_pred $\times$ log_freq	rel_freq $\times$ log_freq
OLMo-2-7B	Full context	0.628 [0.621, 0.634]	0.829 [0.806, 0.853]	-0.014 [-0.016, -0.011]	0.097 [0.090, 0.105]
	Attention-zeroed	0.669 [0.663, 0.676]	0.500 [0.478, 0.523]	-0.013 [-0.016, -0.011]	0.046 [0.038, 0.054]
Llama-1.3B	Full context	0.496 [0.489, 0.503]	1.128 [1.103, 1.153]	-0.010 [-0.013, -0.008]	0.165 [0.156, 0.174]
	Attention-zeroed	0.560 [0.553, 0.567]	0.590 [0.565, 0.616]	-0.017 [-0.020, -0.014]	0.059 [0.050, 0.068]
Llama-3-8B	Full context	0.607 [0.601, 0.614]	0.903 [0.880, 0.925]	-0.015 [-0.018, -0.013]	0.124 [0.116, 0.132]
	Attention-zeroed	0.640 [0.633, 0.647]	0.499 [0.475, 0.523]	-0.016 [-0.018, -0.013]	0.052 [0.045, 0.060]

Figure 7 shows the two moderation terms across models and conditions. The pattern is uniform across all sixteen: $\text{rel_freq} \times \log_freq$ is credibly positive in every model under full context and much smaller under attention-zeroing, while $\text{y_pred} \times \log_freq$ is credibly negative in every large-scale model under both conditions.

G Corpus Frequency Distributions

The figure below compares the frequency distributions of the approximately 47,000 corpus binomials in the BabyLM training corpus (150M tokens, 20 epochs) and the Pile, illustrating the stark difference in lexical coverage between the two training regimes. BabyLM’s distribution is dominated by singleton binomials (70.7% appear exactly once), while Pile frequencies span several orders of magnitude.

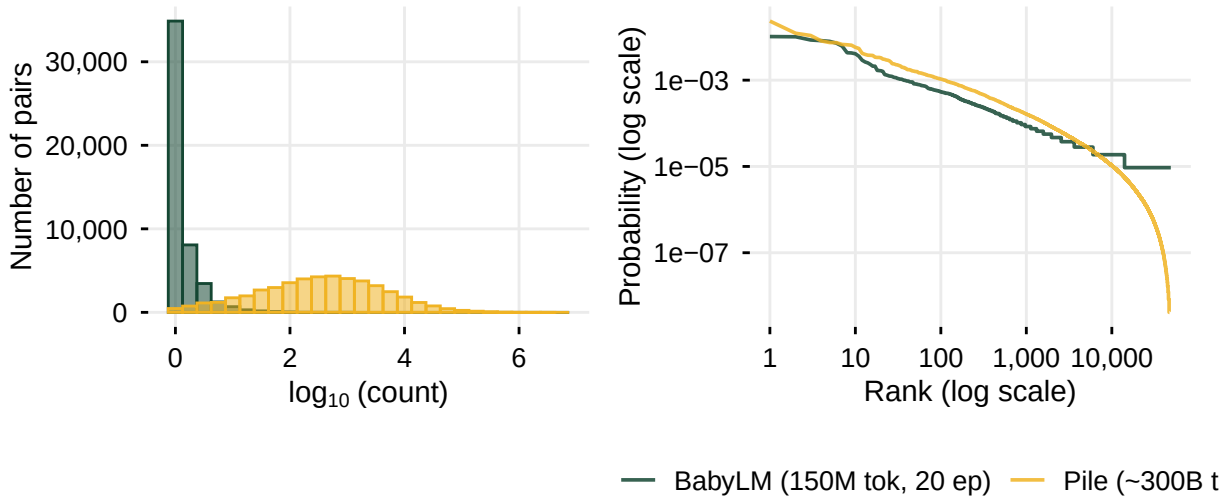


Figure 8: Frequency distributions for the ~47K corpus binomials in the BabyLM training corpus (150M tokens, 20 epochs) and the Pile. Left: histogram of $\log_{10}$ raw counts. Right: rank-frequency plot treating counts as a probability distribution (probability = count / $\sum$ counts); both axes log-scaled. BabyLM’s distribution is heavily concentrated at a single occurrence per pair (70.7% of pairs appear exactly once), whereas Pile frequencies span several orders of magnitude with a median of 365 occurrences per pair.

The BabyLM training corpus is a 150M-token collection of child-directed and informal text (CHILDES, BNC Spoken, children’s books, OpenSubtitles, Wikipedia), trained for 20 epochs. The Pile is a ~300B-token web-scraped corpus used to train the Pythia family. For each of the ~47K corpus binomials, we extracted 3-gram counts (i.e., the frequency of “word₁ and word₂” and its reverse) from each corpus via infini-gram (Liu et al., 2024). As Figure 8 shows, the BabyLM corpus produces a highly degenerate frequency distribution: 70.7% of binomials appear exactly once, 87.4% appear at most twice, and 98.2% appear at most ten times, with a maximum of 1,095 occurrences. The Pile distribution is far richer: the median binomial appears 365 times, the 25th percentile is 62 occurrences, and the 99th percentile exceeds 77,000.

In absolute training exposure terms (accounting for BabyLM’s 20 epochs), the most frequent BabyLM binomials are seen at most ~21,900 times total across all training steps, while the equivalent Pile binomials are seen over 1.5 million times in a single pass. Even the median Pile binomial (365 occurrences per pass) substantially exceeds BabyLM’s total exposure for a typical singleton binomial ($1 \times 20 = 20$ exposures). This gap in absolute exposure is why the two corpora cannot be placed on a common developmental axis, and why the frequency range available for the moderation analysis differs so sharply between them.

H Experiment 1: Supplementary Results

This appendix provides supplementary materials for Experiment 1 (§3), which examined final-layer probe performance for the three BabyLM models. Subsections report full R^2 tables with their label-shuffle controls, and a word-level replication. Coefficients for the frequency-dependent preference regression are reported together for all models in Appendix F.

H.1 Probe and Control R^2

Probe R^2 at the final hidden layer for the BabyLM models, at the binomial level by attention condition, alongside the control probe fit to randomly permuted labels on the same folds. Selectivity is the difference

between the two. Control values near zero indicate that the probe recovers linguistic structure and not label artifacts or probe capacity. Each control is reported beside the R^2 it corresponds to.

Table 5: Probe R^2 at the final hidden layer for BabyLM models, at the binomial level by attention condition, with the label-shuffle control and selectivity (R^2 minus control). B = binomial-held-out split; W = word-held-out split. SD across 10 folds in parentheses. Control values carry an extra decimal place because they sit near zero.

Model	Condition	Split	R^2	Control R^2	Selectivity
BabyLM-125M	Full context	B	0.314 (0.010)	0.0001 (0.0001)	0.314
		W	0.207 (0.023)	0.0002 (0.0002)	0.207
	Attention-zeroed	B	0.244 (0.025)	0.0000 (0.0000)	0.244
		W	0.143 (0.027)	0.0005 (0.0005)	0.142
BabyLM-350M	Full context	B	0.343 (0.019)	0.0000 (0.0000)	0.343
		W	0.225 (0.018)	0.0002 (0.0002)	0.225
	Attention-zeroed	B	0.301 (0.019)	0.0000 (0.0000)	0.301
		W	0.152 (0.028)	0.0001 (0.0001)	0.152
BabyLM-1.3B	Full context	B	0.294 (0.004)	0.0000 (0.0001)	0.294
		W	0.160 (0.022)	0.0002 (0.0002)	0.159
	Attention-zeroed	B	0.263 (0.012)	0.0000 (0.0000)	0.263
		W	0.121 (0.025)	0.0001 (0.0001)	0.121

H.2 Cross-Model Agreement on Novel Binomials

A binomial absent from the training corpus cannot have a retrieved preference, but the model still assigns one, and that value is some deterministic function of the two words’ representations. Such a function could reflect systematic knowledge, or it could be an arbitrary consequence of a particular set of weights. Probe generalization cannot distinguish these, because both are systematic functions of the input and both transfer to held-out words. Agreement between independently trained models can: an arbitrary function of one model’s weights has no reason to match another’s.

We therefore correlate y_{true} across models on identical binomials, separately for the attested and novel sets and for both attention conditions. As a baseline we shuffle one model’s values, which removes any pairing between models while preserving both distributions.

The three BabyLM models are the decisive comparison. They share a training corpus and its ordering but differ in capacity, and they are the only models for which the novel binomials are genuinely absent from training (Section 4, Methods). On novel binomials they agree at $r = 0.45$ under full context and $r = 0.41$ under attention-zeroing, against a shuffled baseline of $r \approx 0.00$. In other words, preferences for binomials the models never encountered are reproducible across models of different sizes trained on the same data. They are therefore not arbitrary.

Because a Pearson correlation can in principle be carried by a minority of high-magnitude items, we checked whether the agreement is concentrated in strong preferences while the bulk of binomials agree at chance. It is not. Preferences on novel binomials are not weak to begin with: median $|y_{\text{true}}|$ ranges from 1.79 to 3.21 across the three models, so the preferred ordering is between roughly six and twenty-five times more likely, and only 9% to 17% of binomials fall below 0.5.

Rank correlations track the Pearson values closely (0.371–0.406 for BabyLM on novel binomials), and the models order the same binomial the same way on 60% to 70% of items against a chance rate of 50%, a statistic that ignores magnitude and weights every binomial equally. Stratifying by preference strength, using the third model’s $|y_{\text{true}}|$ so that the bins are independent of the two series being correlated, agreement is graded but present throughout: in the weakest quintile $r = 0.30$ under full context and 0.28 under attention-zeroing, with directional agreement near 59% in both, rising to $r = 0.61$ and 73% in the strongest. That gradient is itself informative, since agreement increasing with preference strength is the signature of a signal measured with noise rather than of an arbitrary function of a particular set of weights.

Two further patterns appear in Table 6. Agreement within a family is roughly constant across all four cells, near $r = 0.68$, whereas agreement between families falls from 0.56 on attested binomials under full context to 0.28 on novel binomials under attention-zeroing. What the models share thus survives novelty when the training corpus is shared and degrades when it is not, as expected if the preference is

driven by distributional structure in the training data. This also rules out the shared sentence frame as the source of the agreement: the frame is identical for all sixteen models, so it cannot produce a gap between within-family and between-family agreement.

Table 6: Cross-model agreement (Pearson r on $y_{exttrue}$) for identical binomials, by item set and attention condition. Shuffled = one model’s values permuted. Within-family averages over model pairs from the same family; BabyLM is broken out because its novel binomials are genuinely unseen.

Set	Condition	Scope	r	Shuffled
Attested	Attention-zeroed	All	0.470	-0.003
		Within family	0.680	-0.003
		Between family	0.431	-0.003
		Within BabyLM	0.508	-0.003
	Full context	All	0.583	-0.003
		Within family	0.721	-0.003
		Between family	0.557	-0.003
		Within BabyLM	0.553	-0.003
Novel	Attention-zeroed	All	0.346	0.000
		Within family	0.683	0.000
		Between family	0.283	0.000
		Within BabyLM	0.408	0.000
	Full context	All	0.447	-0.002
		Within family	0.679	-0.002
		Between family	0.403	-0.002
		Within BabyLM	0.452	-0.002

H.3 Word-Level Replication

We trained the MLP probe using word-level representations for the three BabyLM models to verify that the preference signal is not an artifact of mean pooling across the full binomial phrase. Under this condition the probe receives a concatenation of two word vectors extracted from isolated sentence contexts (“*the bread*”, “*the butter*”), so it cannot rely on any co-occurrence signal from the two words’ shared sentence context.

Binomial-held-out R^2 ranges from 0.31 to 0.33, within a few percentage points of the corresponding binomial-level values (0.29–0.34). The generalization pattern is preserved: word-held-out R^2 is consistently lower than binomial-held-out R^2 for all three models (word-held-out 0.18–0.20; binomial-minus-word gap 0.127–0.129), confirming that the lexical generalization cost seen at the binomial level is a property of the preference encoding itself rather than an artifact of the mean-pooled representation. The frequency-dependent preference regression was fit at the binomial level only, so it has no word-level counterpart. Full results are in Table 7.

Table 7: Word-level probe R^2 and selectivity (main – control R^2) for BabyLM models (final layer). B = binomial-held-out split; W = word-held-out split. SD across 10 folds in parentheses.

Model	Mode	Split	R ²	Ctrl	Sel.
BabyLM-125M	Word-level	B	0.312 (0.012)	0.000 (0.000)	0.312
		W	0.185 (0.015)	0.000 (0.000)	0.184
B		0.326 (0.010)	0.000 (0.000)	0.326	
W		0.199 (0.016)	0.000 (0.000)	0.198	
B		0.306 (0.003)	0.000 (0.000)	0.306	
W		0.177 (0.023)	0.000 (0.000)	0.177	

I Experiment 2: Supplementary Results

This appendix provides supplementary materials for Experiment 2 (§4), which examined final-layer representations across all 16 models from five families. Subsections report full R^2 tables with their label-shuffle controls, and a word-level replication. Coefficients for the frequency-dependent preference regression are in Appendix F.

I.1 Probe and Control R^2

Probe R^2 for all 16 models at the final hidden layer, binomial-level input mode, by attention condition and cross-validation split, alongside the label-shuffle control fit on the same folds and the resulting selectivity. B = binomial-held-out; W = word-held-out. SD across 10 folds in parentheses.

Table 8: Probe R^2 at the final hidden layer for all 16 models, at the binomial level by attention condition, with the label-shuffle control and selectivity (R^2 minus control). B = binomial-held-out split; W = word-held-out split. SD across 10 folds in parentheses. Control values carry an extra decimal place because they sit near zero.

Model	Condition	Split	R^2	Control R^2	Selectivity
BabyLM-125M	Full context	B	0.314 (0.010)	0.0001 (0.0001)	0.314
		W	0.207 (0.023)	0.0002 (0.0002)	0.207
	Attention-zeroed	B	0.244 (0.025)	0.0000 (0.0000)	0.244
		W	0.143 (0.027)	0.0005 (0.0005)	0.142
BabyLM-350M	Full context	B	0.343 (0.019)	0.0000 (0.0000)	0.343
		W	0.225 (0.018)	0.0002 (0.0002)	0.225
	Attention-zeroed	B	0.301 (0.019)	0.0000 (0.0000)	0.301
		W	0.152 (0.028)	0.0001 (0.0001)	0.152
BabyLM-1.3B	Full context	B	0.294 (0.004)	0.0000 (0.0001)	0.294
		W	0.160 (0.022)	0.0002 (0.0002)	0.159
	Attention-zeroed	B	0.263 (0.012)	0.0000 (0.0000)	0.263
		W	0.121 (0.025)	0.0001 (0.0001)	0.121
GPT-2	Full context	B	0.648 (0.004)	0.0000 (0.0000)	0.648
		W	0.601 (0.033)	0.0005 (0.0006)	0.600
	Attention-zeroed	B	0.786 (0.006)	0.0000 (0.0000)	0.786
		W	0.718 (0.024)	0.0001 (0.0001)	0.718
GPT-2-M	Full context	B	0.533 (0.005)	0.0000 (0.0000)	0.533
		W	0.453 (0.028)	0.0002 (0.0002)	0.453
	Attention-zeroed	B	0.756 (0.016)	0.0000 (0.0000)	0.756
		W	0.670 (0.036)	0.0002 (0.0003)	0.669
GPT-2-L	Full context	B	0.519 (0.006)	0.0000 (0.0000)	0.519
		W	0.449 (0.036)	0.0003 (0.0004)	0.448
	Attention-zeroed	B	0.668 (0.006)	0.0000 (0.0000)	0.668
		W	0.571 (0.020)	0.0004 (0.0005)	0.571
GPT-2-XL	Full context	B	0.487 (0.007)	0.0000 (0.0000)	0.487
		W	0.420 (0.033)	0.0002 (0.0003)	0.420
	Attention-zeroed	B	0.726 (0.008)	0.0000 (0.0000)	0.726
		W	0.641 (0.034)	0.0003 (0.0004)	0.641
Pythia-160M	Full context	B	0.587 (0.005)	0.0000 (0.0000)	0.587
		W	0.522 (0.021)	0.0003 (0.0003)	0.522
	Attention-zeroed	B	0.604 (0.005)	0.0001 (0.0000)	0.603
		W	0.527 (0.022)	0.0007 (0.0005)	0.526
Pythia-410M	Full context	B	0.532 (0.004)	0.0000 (0.0000)	0.532
		W	0.461 (0.035)	0.0003 (0.0003)	0.461
	Attention-zeroed	B	0.552 (0.006)	0.0000 (0.0000)	0.552
		W	0.477 (0.019)	0.0003 (0.0005)	0.476
Pythia-1B	Full context	B	0.554 (0.007)	0.0000 (0.0000)	0.554
		W	0.488 (0.029)	0.0002 (0.0003)	0.487
	Attention-zeroed	B	0.567 (0.005)	0.0000 (0.0000)	0.566
		W	0.496 (0.027)	0.0001 (0.0001)	0.495
OLMo-1B	Full context	B	0.575 (0.005)	0.0000 (0.0000)	0.575
		W	0.505 (0.021)	0.0003 (0.0004)	0.505
	Attention-zeroed	B	0.597 (0.008)	0.0000 (0.0000)	0.597
		W	0.514 (0.021)	0.0003 (0.0004)	0.514

(continued)

Model	Condition	Split	R ²	Control R ²	Selectivity
OLMo-7B	Full context	B	0.621 (0.008)	0.0000 (0.0001)	0.621
		W	0.561 (0.023)	0.0004 (0.0005)	0.561
	Attention-zeroed	B	0.658 (0.008)	0.0000 (0.0000)	0.658
		W	0.580 (0.027)	0.0002 (0.0002)	0.580
OLMo-2-1B	Full context	B	0.557 (0.007)	0.0000 (0.0000)	0.556
		W	0.488 (0.017)	0.0001 (0.0003)	0.488
	Attention-zeroed	B	0.583 (0.005)	0.0000 (0.0000)	0.583
		W	0.518 (0.021)	0.0004 (0.0003)	0.518
OLMo-2-7B	Full context	B	0.614 (0.005)	0.0000 (0.0000)	0.614
		W	0.554 (0.022)	0.0004 (0.0004)	0.554
	Attention-zeroed	B	0.642 (0.006)	0.0000 (0.0000)	0.642
		W	0.570 (0.012)	0.0003 (0.0004)	0.569
Llama-1.3B	Full context	B	0.433 (0.008)	0.0000 (0.0000)	0.433
		W	0.373 (0.035)	0.0003 (0.0004)	0.373
	Attention-zeroed	B	0.443 (0.011)	0.0000 (0.0000)	0.443
		W	0.370 (0.040)	0.0002 (0.0003)	0.370
Llama-3-8B	Full context	B	0.545 (0.008)	0.0000 (0.0000)	0.545
		W	0.510 (0.035)	0.0007 (0.0006)	0.510
	Attention-zeroed	B	0.521 (0.016)	0.0000 (0.0000)	0.521
		W	0.470 (0.024)	0.0002 (0.0002)	0.470

I.2 Word-Level Replication

Across all 16 models, word-level probing reproduces the main findings from Experiment 2. Binomial-held-out R² is slightly lower than the corresponding binomial-level values (by roughly 5–10 percentage points for most large-scale models), but rank order across families and the generalization pattern (binomial-held-out > word-held-out) are preserved in every case. For GPT-2, binomial-held-out R² under word-level probing ranges from 0.46 to 0.62 (vs. 0.49–0.65 at the binomial level); for Pythia, 0.51–0.57 (vs. 0.53–0.59); for OLMo, 0.52–0.57 (vs. 0.56–0.62); for Llama, 0.36–0.45 (vs. 0.43–0.55). The generalization gaps (binomial-held-out minus word-held-out) range from 0.044 to 0.073 across large-scale models, closely matching the corresponding binomial-level gaps (0.035–0.080). Full results are in Table 9.

Table 9: Word-level probe R² and selectivity (main – control R²) for all 16 models (final layer). B = binomial-held-out split; W = word-held-out split. SD across 10 folds in parentheses.

Family	Model	Mode	Split	R ²	Ctrl	Sel.
BabyLM	BabyLM-125M	B		0.312 (0.012)	0.000 (0.000)	0.312
		W		0.185 (0.015)	0.000 (0.000)	0.184
	BabyLM-350M	B		0.326 (0.010)	0.000 (0.000)	0.326
		W		0.199 (0.016)	0.000 (0.000)	0.198
GPT-2	BabyLM-1.3B	B		0.306 (0.003)	0.000 (0.000)	0.306
		W		0.177 (0.023)	0.000 (0.000)	0.177
	GPT-2	B		0.618 (0.003)	0.000 (0.000)	0.618
		W		0.574 (0.032)	0.000 (0.000)	0.573
	GPT-2-M	B		0.521 (0.006)	0.000 (0.000)	0.521
		W		0.448 (0.022)	0.000 (0.000)	0.447
	GPT-2-L	B		0.487 (0.005)	0.000 (0.000)	0.487
		W		0.425 (0.027)	0.000 (0.000)	0.425
Llama	GPT-2-XL	B		0.458 (0.005)	0.000 (0.000)	0.458
		W		0.391 (0.034)	0.000 (0.000)	0.390
	Llama-1.3B	B		0.357 (0.010)	0.000 (0.000)	0.357
		W		0.293 (0.036)	0.000 (0.001)	0.292
	Llama-3-8B	B		0.448 (0.010)	0.000 (0.000)	0.448
		W		0.402 (0.027)	0.000 (0.001)	0.401
	OLMo-1B	B		0.537 (0.004)	0.000 (0.000)	0.537
		W		0.471 (0.021)	0.000 (0.000)	0.471
OLMo	OLMo-7B	B		0.566 (0.004)	0.000 (0.000)	0.566
		W		0.503 (0.026)	0.000 (0.000)	0.502
	OLMo-2-1B	B		0.520 (0.008)	0.000 (0.000)	0.520
		W		0.453 (0.018)	0.000 (0.000)	0.453
	OLMo-2-7B	B		0.549 (0.008)	0.000 (0.000)	0.549
		W		0.478 (0.023)	0.000 (0.001)	0.478

(continued)

Family	Model	Mode	Split	R ²	Ctrl	Sel.
Pythia	Pythia-160M		B	0.570 (0.006)	0.000 (0.000)	0.570
			W	0.507 (0.021)	0.000 (0.001)	0.507
	Pythia-410M		B	0.514 (0.005)	0.000 (0.000)	0.514
			W	0.458 (0.025)	0.000 (0.000)	0.457
	Pythia-1B		B	0.523 (0.006)	0.000 (0.000)	0.523
			W	0.454 (0.024)	0.000 (0.000)	0.454

J Experiment 3: Supplementary Results

This appendix provides supplementary materials for Experiment 3 (§5), which tracked probe R^2 and the frequency-dependent preference regression across training checkpoints for BabyLM and Pythia. Subsections report full numerical checkpoint tables and a word-level replication of the probe results.

J.1 Checkpoint Probe Tables

Full numerical results for the Experiment 3 checkpoint analysis. Tables report probe R^2 (binomial-held-out and word-held-out) and the coefficients of Equation 1 for BabyLM and Pythia across training checkpoints, for the binomial-level (mean-pooled) representation under full-context and attention-zeroed conditions.

Table 10: BabyLM checkpoint probe R^2 :full context, binomial-level. B = binomial-held-out; W = word-held-out. SD in parentheses.

Model	Step	Tokens	Split	R^2	Ctrl	Sel.
BabyLM-125M	24	20 M	B	0.671 (0.022)	0.000 (0.000)	0.671
		20 M	W	0.434 (0.054)	0.000 (0.000)	0.434
	48	39 M	B	0.579 (0.041)	0.000 (0.000)	0.579
		39 M	W	0.353 (0.045)	0.000 (0.001)	0.353
	144	118 M	B	0.620 (0.023)	0.000 (0.000)	0.620
		118 M	W	0.439 (0.035)	0.000 (0.000)	0.439
	384	315 M	B	0.610 (0.016)	0.000 (0.000)	0.610
		315 M	W	0.476 (0.038)	0.000 (0.000)	0.476
	912	747 M	B	0.505 (0.034)	0.000 (0.000)	0.505
		747 M	W	0.346 (0.055)	0.000 (0.000)	0.346
	2280	1868 M	B	0.342 (0.013)	0.000 (0.000)	0.342
		1868 M	W	0.227 (0.030)	0.000 (0.000)	0.227
BabyLM-350M	48	20 M	B	0.684 (0.031)	0.000 (0.000)	0.684
		20 M	W	0.582 (0.041)	0.000 (0.000)	0.582
	96	39 M	B	0.640 (0.050)	0.000 (0.000)	0.640
		39 M	W	0.575 (0.029)	0.000 (0.000)	0.575
	288	118 M	B	0.661 (0.027)	0.000 (0.000)	0.661
		118 M	W	0.592 (0.033)	0.000 (0.000)	0.592
	768	315 M	B	0.640 (0.024)	0.000 (0.000)	0.640
		315 M	W	0.532 (0.040)	0.000 (0.000)	0.532
	1824	747 M	B	0.503 (0.026)	0.000 (0.000)	0.503
		747 M	W	0.361 (0.048)	0.000 (0.000)	0.361
	4560	1868 M	B	0.382 (0.013)	0.000 (0.000)	0.382
		1868 M	W	0.240 (0.025)	0.000 (0.000)	0.239
BabyLM-1.3B	97	20 M	B	0.718 (0.020)	0.000 (0.000)	0.718
		20 M	W	0.527 (0.045)	0.001 (0.001)	0.527
	194	40 M	B	0.635 (0.029)	0.000 (0.000)	0.635
		40 M	W	0.434 (0.064)	0.001 (0.000)	0.433
	582	119 M	B	0.653 (0.017)	0.000 (0.000)	0.653
		119 M	W	0.553 (0.035)	0.000 (0.000)	0.553
	1455	298 M	B	0.555 (0.023)	0.000 (0.000)	0.555
		298 M	W	0.426 (0.044)	0.000 (0.000)	0.426
	3686	755 M	B	0.394 (0.011)	0.000 (0.000)	0.394
		755 M	W	0.248 (0.020)	0.000 (0.000)	0.247
	9021	1848 M	B	0.315 (0.007)	0.000 (0.000)	0.315
		1848 M	W	0.180 (0.024)	0.000 (0.000)	0.179

Table 11: BabyLM checkpoint probe R²:attention-zeroed, binomial-level.

Model	Step	Tokens	Split	R ²	Ctrl	Sel.
BabyLM-125M	24	20 M	B	0.799 (0.019)	0.000 (0.000)	0.799
		20 M	W	0.712 (0.041)	0.000 (0.000)	0.711
	48	39 M	B	0.770 (0.012)	0.000 (0.000)	0.770
		39 M	W	0.659 (0.041)	0.001 (0.001)	0.658
	144	118 M	B	0.691 (0.017)	0.000 (0.000)	0.691
		118 M	W	0.588 (0.041)	0.000 (0.000)	0.588
	384	315 M	B	0.612 (0.013)	0.000 (0.000)	0.612
		315 M	W	0.512 (0.032)	0.000 (0.000)	0.511
	912	747 M	B	0.458 (0.015)	0.000 (0.000)	0.458
		747 M	W	0.320 (0.047)	0.001 (0.001)	0.319
	2280	1868 M	B	0.283 (0.019)	0.000 (0.000)	0.283
		1868 M	W	0.150 (0.029)	0.000 (0.001)	0.150
BabyLM-350M	48	20 M	B	0.784 (0.011)	0.000 (0.000)	0.784
		20 M	W	0.738 (0.021)	0.000 (0.001)	0.737
	96	39 M	B	0.847 (0.011)	0.000 (0.000)	0.847
		39 M	W	0.813 (0.022)	0.000 (0.000)	0.813
	288	118 M	B	0.730 (0.047)	0.000 (0.000)	0.730
		118 M	W	0.711 (0.025)	0.000 (0.000)	0.710
	768	315 M	B	0.598 (0.040)	0.000 (0.000)	0.598
		315 M	W	0.515 (0.041)	0.000 (0.000)	0.515
	1824	747 M	B	0.477 (0.019)	0.000 (0.000)	0.477
		747 M	W	0.318 (0.042)	0.000 (0.000)	0.318
	4560	1868 M	B	0.327 (0.016)	0.000 (0.000)	0.327
		1868 M	W	0.167 (0.031)	0.000 (0.000)	0.167
BabyLM-1.3B	97	20 M	B	0.803 (0.016)	0.000 (0.000)	0.803
		20 M	W	0.755 (0.025)	0.000 (0.001)	0.755
	194	40 M	B	0.734 (0.025)	0.000 (0.000)	0.734
		40 M	W	0.665 (0.033)	0.000 (0.001)	0.665
	582	119 M	B	0.672 (0.015)	0.000 (0.000)	0.672
		119 M	W	0.602 (0.031)	0.000 (0.000)	0.601
	1455	298 M	B	0.527 (0.014)	0.000 (0.000)	0.527
		298 M	W	0.410 (0.060)	0.000 (0.000)	0.410
	3686	755 M	B	0.347 (0.019)	0.000 (0.000)	0.347
		755 M	W	0.193 (0.020)	0.000 (0.000)	0.193
	9021	1848 M	B	0.268 (0.007)	0.000 (0.000)	0.268
		1848 M	W	0.131 (0.026)	0.000 (0.000)	0.131

Table 12: Pythia checkpoint probe R²:full context, binomial-level.

Model	Step	Tokens	Split	R ²	Ctrl	Sel.
Pythia-160M	16	34 M	B	0.681 (0.005)	0.000 (0.000)	0.681
		34 M	W	0.380 (0.032)	0.000 (0.000)	0.380
	32	67 M	B	0.695 (0.004)	0.000 (0.000)	0.695
		67 M	W	0.398 (0.034)	0.000 (0.000)	0.398
	64	134 M	B	0.670 (0.010)	0.000 (0.000)	0.670
		134 M	W	0.415 (0.034)	0.000 (0.000)	0.415
	256	537 M	B	0.799 (0.008)	0.000 (0.000)	0.799
		537 M	W	0.649 (0.035)	0.000 (0.000)	0.648
	512	1074 M	B	0.770 (0.006)	0.000 (0.000)	0.770
		1074 M	W	0.644 (0.019)	0.000 (0.000)	0.644
	1000	2097 M	B	0.715 (0.007)	0.000 (0.000)	0.715
		2097 M	W	0.632 (0.019)	0.000 (0.001)	0.632

(continued)

Model	Step	Tokens	Split	R ²	Ctrl	Sel.
Pythia-410M	16	34 M	B	0.692 (0.007)	0.000 (0.000)	0.691
		34 M	W	0.405 (0.035)	0.000 (0.000)	0.405
	32	67 M	B	0.700 (0.007)	0.000 (0.000)	0.700
		67 M	W	0.426 (0.035)	0.000 (0.000)	0.426
	64	134 M	B	0.687 (0.007)	0.000 (0.000)	0.687
		134 M	W	0.443 (0.039)	0.000 (0.000)	0.443
	256	537 M	B	0.814 (0.008)	0.000 (0.000)	0.814
		537 M	W	0.719 (0.019)	0.000 (0.000)	0.719
	512	1074 M	B	0.801 (0.007)	0.000 (0.000)	0.801
		1074 M	W	0.714 (0.018)	0.000 (0.000)	0.713
	1000	2097 M	B	0.758 (0.002)	0.000 (0.000)	0.757
		2097 M	W	0.690 (0.011)	0.000 (0.000)	0.689
Pythia-1B	16	34 M	B	0.737 (0.004)	0.000 (0.000)	0.737
		34 M	W	0.440 (0.026)	0.000 (0.000)	0.440
	32	67 M	B	0.755 (0.004)	0.000 (0.000)	0.755
		67 M	W	0.470 (0.033)	0.000 (0.000)	0.470
	64	134 M	B	0.747 (0.007)	0.000 (0.000)	0.747
		134 M	W	0.500 (0.018)	0.000 (0.001)	0.499
	256	537 M	B	0.864 (0.006)	0.000 (0.000)	0.864
		537 M	W	0.754 (0.025)	0.000 (0.000)	0.754
	512	1074 M	B	0.781 (0.005)	0.000 (0.000)	0.781
		1074 M	W	0.664 (0.028)	0.000 (0.001)	0.664
	1000	2097 M	B	0.764 (0.005)	0.000 (0.000)	0.764
		2097 M	W	0.683 (0.016)	0.000 (0.000)	0.683

Table 13: Pythia checkpoint probe R²:attention-zeroed, binomial-level.

Model	Step	Tokens	Split	R ²	Ctrl	Sel.
Pythia-160M	16	34 M	B	0.692 (0.009)	0.000 (0.000)	0.692
		34 M	W	0.326 (0.058)	0.000 (0.000)	0.326
	32	67 M	B	0.720 (0.009)	0.000 (0.000)	0.720
		67 M	W	0.357 (0.062)	0.000 (0.000)	0.357
	64	134 M	B	0.677 (0.014)	0.000 (0.000)	0.677
		134 M	W	0.309 (0.057)	0.000 (0.000)	0.309
	256	537 M	B	0.749 (0.014)	0.000 (0.000)	0.749
		537 M	W	0.503 (0.043)	0.000 (0.000)	0.503
	512	1074 M	B	0.752 (0.008)	0.000 (0.000)	0.752
		1074 M	W	0.579 (0.031)	0.000 (0.000)	0.578
	1000	2097 M	B	0.716 (0.007)	0.000 (0.000)	0.716
		2097 M	W	0.611 (0.020)	0.001 (0.001)	0.611
Pythia-410M	16	34 M	B	0.731 (0.005)	0.000 (0.000)	0.731
		34 M	W	0.378 (0.030)	0.000 (0.000)	0.377
	32	67 M	B	0.711 (0.015)	0.000 (0.000)	0.711
		67 M	W	0.352 (0.019)	0.000 (0.000)	0.351
	64	134 M	B	0.676 (0.014)	0.000 (0.000)	0.676
		134 M	W	0.334 (0.053)	0.000 (0.000)	0.334
	256	537 M	B	0.745 (0.017)	0.000 (0.000)	0.745
		537 M	W	0.505 (0.036)	0.000 (0.000)	0.505
	512	1074 M	B	0.769 (0.020)	0.000 (0.000)	0.769
		1074 M	W	0.618 (0.023)	0.000 (0.000)	0.618
	1000	2097 M	B	0.751 (0.009)	0.000 (0.000)	0.751
		2097 M	W	0.659 (0.015)	0.000 (0.000)	0.659

(continued)

Model	Step	Tokens	Split	R ²	Ctrl	Sel.
Pythia-1B	16	34 M	B	0.797 (0.011)	0.000 (0.000)	0.797
		34 M	W	0.494 (0.033)	0.000 (0.000)	0.493
	32	67 M	B	0.804 (0.010)	0.000 (0.000)	0.804
		67 M	W	0.495 (0.028)	0.000 (0.000)	0.494
	64	134 M	B	0.749 (0.015)	0.000 (0.000)	0.749
		134 M	W	0.360 (0.063)	0.000 (0.000)	0.360
	256	537 M	B	0.827 (0.011)	0.000 (0.000)	0.827
		537 M	W	0.622 (0.021)	0.001 (0.001)	0.621
	512	1074 M	B	0.773 (0.010)	0.000 (0.000)	0.773
		1074 M	W	0.589 (0.027)	0.000 (0.000)	0.588
	1000	2097 M	B	0.760 (0.007)	0.000 (0.000)	0.760
		2097 M	W	0.660 (0.017)	0.000 (0.000)	0.660

Table 14: Frequency-dependent preference regression (Equation 1) at every training checkpoint, both attention conditions. y_{true} and y_{pred} are z-scored within each fit; rel_freq is a proportion from -0.5 to $+0.5$ and $\log_freq$ is centered but not scaled, so both mean the same thing in every corpus. 95% credible intervals in brackets.

Model	Step	Condition	y_{pred}	rel_freq	$y_{\text{pred}} \times \log_freq$	$\text{rel_freq} \times \log_freq$
BabyLM-125M	24	Full context	0.625 [0.618, 0.632]	-0.020 [-0.035, -0.006]	0.033 [0.021, 0.045]	0.003 [-0.026, 0.032]
	48	Full context	0.613 [0.606, 0.619]	-0.035 [-0.050, -0.020]	0.017 [0.005, 0.029]	-0.004 [-0.034, 0.026]
	144	Full context	0.577 [0.570, 0.584]	-0.007 [-0.022, 0.008]	-0.014 [-0.025, -0.003]	0.014 [-0.015, 0.043]
	384	Full context	0.502 [0.495, 0.510]	0.139 [0.123, 0.155]	-0.009 [-0.022, 0.003]	0.067 [0.038, 0.098]
	912	Full context	0.472 [0.465, 0.480]	0.248 [0.231, 0.265]	-0.002 [-0.014, 0.010]	0.151 [0.121, 0.182]
	2280	Full context	0.428 [0.420, 0.436]	0.463 [0.447, 0.479]	-0.003 [-0.015, 0.009]	0.338 [0.307, 0.367]
	24	Attention-zeroed	0.824 [0.819, 0.829]	-0.010 [-0.021, 0.001]	0.018 [0.009, 0.026]	0.011 [-0.009, 0.032]
	48	Attention-zeroed	0.809 [0.803, 0.814]	-0.009 [-0.020, 0.002]	0.019 [0.011, 0.028]	0.012 [-0.010, 0.034]
	144	Attention-zeroed	0.648 [0.641, 0.655]	-0.001 [-0.015, 0.014]	0.001 [-0.010, 0.013]	0.017 [-0.011, 0.044]
	384	Attention-zeroed	0.502 [0.494, 0.510]	0.157 [0.141, 0.173]	-0.023 [-0.034, -0.012]	0.074 [0.043, 0.104]
	912	Attention-zeroed	0.417 [0.409, 0.425]	0.263 [0.246, 0.280]	-0.014 [-0.027, -0.002]	0.139 [0.109, 0.170]
	2280	Attention-zeroed	0.396 [0.388, 0.404]	0.395 [0.379, 0.412]	-0.025 [-0.037, -0.014]	0.196 [0.167, 0.225]
BabyLM-350M	48	Full context	0.670 [0.664, 0.677]	-0.027 [-0.041, -0.013]	-0.004 [-0.016, 0.007]	0.008 [-0.020, 0.036]
	96	Full context	0.667 [0.661, 0.674]	-0.020 [-0.034, -0.006]	-0.011 [-0.022, -0.001]	-0.003 [-0.031, 0.024]
	288	Full context	0.675 [0.669, 0.682]	0.003 [-0.011, 0.017]	-0.024 [-0.035, -0.014]	0.019 [-0.008, 0.046]
	768	Full context	0.563 [0.556, 0.571]	0.140 [0.125, 0.155]	-0.014 [-0.025, -0.003]	0.051 [0.022, 0.079]
	1824	Full context	0.512 [0.505, 0.520]	0.252 [0.236, 0.266]	-0.001 [-0.013, 0.011]	0.153 [0.123, 0.182]
	4560	Full context	0.438 [0.430, 0.445]	0.414 [0.397, 0.430]	-0.014 [-0.026, -0.003]	0.302 [0.270, 0.333]
	48	Attention-zeroed	0.811 [0.806, 0.817]	-0.013 [-0.024, -0.001]	-0.004 [-0.013, 0.004]	0.015 [-0.007, 0.038]
	96	Attention-zeroed	0.862 [0.857, 0.866]	-0.008 [-0.018, 0.001]	-0.005 [-0.012, 0.002]	0.009 [-0.009, 0.026]
	288	Attention-zeroed	0.718 [0.711, 0.724]	0.008 [-0.005, 0.021]	-0.047 [-0.057, -0.037]	0.012 [-0.015, 0.039]
	768	Attention-zeroed	0.510 [0.502, 0.518]	0.169 [0.153, 0.185]	-0.028 [-0.040, -0.016]	0.068 [0.038, 0.098]
	1824	Attention-zeroed	0.480 [0.473, 0.488]	0.249 [0.233, 0.265]	-0.014 [-0.026, -0.003]	0.139 [0.109, 0.169]
	4560	Attention-zeroed	0.402 [0.394, 0.409]	0.381 [0.364, 0.398]	-0.032 [-0.042, -0.021]	0.199 [0.169, 0.229]
BabyLM-1.3B	97	Full context	0.625 [0.618, 0.632]	-0.010 [-0.024, 0.005]	-0.008 [-0.020, 0.003]	0.029 [0.000, 0.058]
	194	Full context	0.559 [0.552, 0.567]	-0.008 [-0.023, 0.008]	-0.001 [-0.013, 0.011]	0.026 [-0.004, 0.056]
	582	Full context	0.551 [0.543, 0.559]	0.069 [0.054, 0.084]	-0.009 [-0.020, 0.002]	0.047 [0.019, 0.076]
	1455	Full context	0.498 [0.490, 0.505]	0.216 [0.200, 0.232]	-0.007 [-0.019, 0.005]	0.109 [0.079, 0.138]
	3686	Full context	0.409 [0.402, 0.417]	0.477 [0.460, 0.494]	-0.023 [-0.034, -0.011]	0.340 [0.309, 0.371]
	9021	Full context	0.323 [0.316, 0.331]	0.802 [0.786, 0.818]	-0.018 [-0.030, -0.006]	0.459 [0.430, 0.489]
	97	Attention-zeroed	0.796 [0.791, 0.801]	-0.003 [-0.014, 0.008]	0.004 [-0.004, 0.013]	0.014 [-0.007, 0.036]
	194	Attention-zeroed	0.726 [0.720, 0.732]	0.011 [-0.001, 0.024]	-0.006 [-0.016, 0.004]	0.039 [0.015, 0.064]
	582	Attention-zeroed	0.588 [0.581, 0.595]	0.070 [0.055, 0.085]	-0.019 [-0.030, -0.008]	0.057 [0.030, 0.085]
	1455	Attention-zeroed	0.484 [0.476, 0.492]	0.227 [0.211, 0.243]	-0.016 [-0.028, -0.005]	0.097 [0.067, 0.125]
	3686	Attention-zeroed	0.390 [0.381, 0.397]	0.393 [0.375, 0.410]	-0.037 [-0.049, -0.025]	0.154 [0.123, 0.183]
	9021	Attention-zeroed	0.298 [0.290, 0.306]	0.612 [0.595, 0.629]	-0.025 [-0.037, -0.013]	0.168 [0.139, 0.198]
Pythia-160M	16	Full context	0.528 [0.521, 0.536]	-0.021 [-0.048, 0.006]	0.009 [0.006, 0.012]	0.014 [0.004, 0.023]
	32	Full context	0.570 [0.562, 0.577]	-0.030 [-0.057, -0.003]	0.004 [0.001, 0.007]	0.009 [0.000, 0.018]
	64	Full context	0.557 [0.550, 0.565]	-0.002 [-0.029, 0.024]	0.009 [0.006, 0.013]	0.018 [0.009, 0.027]
	256	Full context	0.686 [0.679, 0.692]	0.242 [0.219, 0.266]	-0.003 [-0.005, -0.000]	0.026 [0.018, 0.034]
	512	Full context	0.669 [0.662, 0.675]	0.420 [0.397, 0.443]	-0.002 [-0.005, 0.001]	0.058 [0.049, 0.066]
	1000	Full context	0.622 [0.615, 0.629]	0.488 [0.464, 0.512]	0.003 [-0.000, 0.005]	0.097 [0.089, 0.105]
	16	Attention-zeroed	0.553 [0.545, 0.560]	0.027 [0.001, 0.054]	-0.006 [-0.009, -0.003]	0.014 [0.005, 0.023]
	32	Attention-zeroed	0.600 [0.593, 0.607]	0.050 [0.025, 0.076]	-0.007 [-0.010, -0.004]	0.010 [0.001, 0.019]
	64	Attention-zeroed	0.582 [0.574, 0.589]	0.058 [0.033, 0.084]	-0.009 [-0.012, -0.006]	0.015 [0.006, 0.024]
	256	Attention-zeroed	0.651 [0.645, 0.658]	0.246 [0.222, 0.270]	-0.008 [-0.011, -0.005]	0.031 [0.022, 0.039]
	512	Attention-zeroed	0.647 [0.641, 0.654]	0.460 [0.437, 0.484]	-0.004 [-0.007, -0.001]	0.049 [0.041, 0.057]
	1000	Attention-zeroed	0.617 [0.610, 0.624]	0.388 [0.362, 0.413]	-0.002 [-0.005, 0.001]	0.045 [0.036, 0.053]

(continued)

Model	Step	Condition	y_pred	rel_freq	y_pred $\times$ log_freq	rel_freq $\times$ log_freq
Pythia-410M	16	Full context	0.552 [0.544, 0.559]	-0.013 [-0.039, 0.014]	0.010 [0.007, 0.013]	0.014 [0.005, 0.024]
	32	Full context	0.575 [0.568, 0.582]	-0.012 [-0.037, 0.014]	0.008 [0.005, 0.011]	0.011 [0.002, 0.020]
	64	Full context	0.581 [0.573, 0.589]	0.042 [0.016, 0.068]	0.009 [0.006, 0.012]	0.021 [0.012, 0.030]
	256	Full context	0.727 [0.721, 0.733]	0.174 [0.152, 0.196]	0.006 [0.003, 0.008]	0.024 [0.016, 0.031]
	512	Full context	0.682 [0.675, 0.688]	0.379 [0.356, 0.402]	-0.004 [-0.007, -0.001]	0.051 [0.043, 0.059]
	1000	Full context	0.638 [0.631, 0.645]	0.488 [0.465, 0.512]	-0.000 [-0.003, 0.003]	0.090 [0.082, 0.098]
	16	Attention-zeroed	0.595 [0.588, 0.603]	0.054 [0.029, 0.080]	-0.004 [-0.007, -0.001]	0.013 [0.004, 0.021]
	32	Attention-zeroed	0.581 [0.574, 0.589]	0.076 [0.050, 0.102]	-0.006 [-0.010, -0.004]	0.015 [0.006, 0.024]
	64	Attention-zeroed	0.565 [0.558, 0.573]	0.098 [0.072, 0.125]	-0.009 [-0.012, -0.006]	0.022 [0.014, 0.032]
	256	Attention-zeroed	0.666 [0.659, 0.673]	0.218 [0.194, 0.242]	-0.005 [-0.008, -0.002]	0.024 [0.016, 0.032]
Pythia-1B	512	Attention-zeroed	0.655 [0.649, 0.662]	0.409 [0.385, 0.433]	-0.003 [-0.006, -0.000]	0.051 [0.043, 0.059]
	1000	Attention-zeroed	0.644 [0.637, 0.651]	0.392 [0.367, 0.415]	-0.002 [-0.004, 0.001]	0.048 [0.040, 0.057]
	16	Full context	0.614 [0.606, 0.621]	-0.018 [-0.044, 0.007]	0.008 [0.005, 0.010]	0.017 [0.009, 0.026]
	32	Full context	0.627 [0.619, 0.634]	-0.002 [-0.026, 0.023]	0.009 [0.006, 0.012]	0.021 [0.013, 0.030]
	64	Full context	0.621 [0.614, 0.628]	0.026 [0.001, 0.050]	0.009 [0.006, 0.012]	0.029 [0.020, 0.038]
	256	Full context	0.775 [0.769, 0.781]	0.264 [0.244, 0.284]	-0.002 [-0.004, 0.001]	0.037 [0.030, 0.044]
	512	Full context	0.674 [0.668, 0.681]	0.409 [0.387, 0.432]	-0.003 [-0.006, -0.000]	0.070 [0.062, 0.078]
	1000	Full context	0.654 [0.647, 0.660]	0.530 [0.507, 0.554]	0.005 [0.002, 0.007]	0.114 [0.106, 0.121]
	16	Attention-zeroed	0.696 [0.690, 0.703]	0.052 [0.028, 0.075]	-0.009 [-0.012, -0.007]	0.009 [0.001, 0.017]
	32	Attention-zeroed	0.710 [0.704, 0.717]	0.057 [0.034, 0.078]	-0.010 [-0.013, -0.008]	0.011 [0.003, 0.019]
Pythia-1B	64	Attention-zeroed	0.628 [0.621, 0.635]	0.100 [0.076, 0.125]	-0.005 [-0.008, -0.002]	0.028 [0.020, 0.037]
	256	Attention-zeroed	0.728 [0.722, 0.734]	0.278 [0.257, 0.299]	-0.008 [-0.011, -0.006]	0.037 [0.029, 0.044]
	512	Attention-zeroed	0.643 [0.636, 0.650]	0.446 [0.422, 0.469]	-0.002 [-0.005, 0.001]	0.064 [0.056, 0.073]
	1000	Attention-zeroed	0.652 [0.645, 0.659]	0.410 [0.387, 0.434]	-0.003 [-0.006, -0.001]	0.060 [0.052, 0.068]

J.2 Word-Level Replication

Word-level probe results across BabyLM and Pythia checkpoints replicate the key patterns from the binomial-level analysis. The generalization gap (binomial-held-out minus word-held-out) is largest at the earliest checkpoints and shrinks over the course of training. For BabyLM, the gap at the first checkpoint ranges from 0.15 to 0.36 across model sizes, narrowing to 0.13–0.14 by the final checkpoint, the same qualitative trajectory observed at the binomial level. Full R^2 results are in Table 15 and Table 16. The frequency-dependent preference regression was fit at the binomial level only, so it has no word-level counterpart here; its checkpoint coefficients are in Table 14.

Table 15: BabyLM checkpoint probe R^2 :full context, word-level. B = binomial-held-out; W = word-held-out. SD in parentheses.

Model	Step	Tokens	Split	R^2	Ctrl	Sel.
BabyLM-125M	24	20 M	B	0.450 (0.056)	0.000 (0.000)	0.450
		20 M	W	0.205 (0.041)	0.000 (0.001)	0.205
	48	39 M	B	0.384 (0.055)	0.000 (0.000)	0.384
		39 M	W	0.158 (0.031)	0.001 (0.001)	0.158
	144	118 M	B	0.355 (0.072)	0.000 (0.000)	0.355
		118 M	W	0.129 (0.035)	0.000 (0.000)	0.129
	384	315 M	B	0.403 (0.052)	0.000 (0.000)	0.403
		315 M	W	0.213 (0.024)	0.000 (0.000)	0.212
	912	747 M	B	0.400 (0.034)	0.000 (0.000)	0.400
		747 M	W	0.231 (0.027)	0.001 (0.001)	0.231
	2280	1868 M	B	0.335 (0.020)	0.000 (0.000)	0.335
		1868 M	W	0.208 (0.017)	0.000 (0.000)	0.208
BabyLM-350M	48	20 M	B	0.340 (0.050)	0.000 (0.000)	0.340
		20 M	W	0.188 (0.036)	0.000 (0.001)	0.187
	96	39 M	B	0.232 (0.027)	0.000 (0.000)	0.232
		39 M	W	0.137 (0.052)	0.000 (0.000)	0.136
	288	118 M	B	0.167 (0.030)	0.000 (0.000)	0.167
		118 M	W	0.092 (0.038)	0.000 (0.000)	0.091
	768	315 M	B	0.372 (0.043)	0.000 (0.000)	0.372
		315 M	W	0.199 (0.041)	0.000 (0.000)	0.199
	1824	747 M	B	0.399 (0.041)	0.000 (0.000)	0.399
		747 M	W	0.246 (0.040)	0.000 (0.000)	0.245
	4560	1868 M	B	0.343 (0.035)	0.000 (0.000)	0.343
		1868 M	W	0.208 (0.017)	0.000 (0.000)	0.207

(continued)

Model	Step	Tokens	Split	R ²	Ctrl	Sel.
BabyLM-1.3B	97	20 M	B	0.570 (0.032)	0.000 (0.000)	0.570
		20 M	W	0.215 (0.052)	0.000 (0.000)	0.214
	194	40 M	B	0.444 (0.037)	0.000 (0.000)	0.444
		40 M	W	0.127 (0.047)	0.000 (0.000)	0.127
	582	119 M	B	0.421 (0.032)	0.000 (0.000)	0.421
		119 M	W	0.187 (0.024)	0.000 (0.000)	0.186
	1455	298 M	B	0.462 (0.026)	0.000 (0.000)	0.462
		298 M	W	0.283 (0.052)	0.000 (0.000)	0.283
	3686	755 M	B	0.394 (0.009)	0.000 (0.000)	0.394
		755 M	W	0.236 (0.019)	0.000 (0.000)	0.235
	9021	1848 M	B	0.320 (0.004)	0.000 (0.000)	0.320
		1848 M	W	0.185 (0.022)	0.000 (0.000)	0.184

Table 16: Pythia checkpoint probe R²:full context, word-level. B = binomial-held-out; W = word-held-out. SD in parentheses.

Model	Step	Tokens	Split	R ²	Ctrl	Sel.
Pythia-160M	16	34 M	B	0.563 (0.009)	0.000 (0.000)	0.562
		34 M	W	0.226 (0.033)	0.000 (0.000)	0.226
	32	67 M	B	0.577 (0.007)	0.000 (0.000)	0.577
		67 M	W	0.238 (0.026)	0.000 (0.000)	0.238
	64	134 M	B	0.565 (0.006)	0.000 (0.000)	0.565
		134 M	W	0.239 (0.033)	0.000 (0.000)	0.239
	256	537 M	B	0.615 (0.013)	0.000 (0.000)	0.615
		537 M	W	0.332 (0.034)	0.000 (0.000)	0.332
	512	1074 M	B	0.689 (0.004)	0.000 (0.000)	0.689
		1074 M	W	0.527 (0.016)	0.000 (0.000)	0.527
	1000	2097 M	B	0.667 (0.006)	0.000 (0.000)	0.667
		2097 M	W	0.562 (0.020)	0.000 (0.001)	0.561
Pythia-410M	16	34 M	B	0.576 (0.006)	0.000 (0.000)	0.576
		34 M	W	0.235 (0.021)	0.000 (0.000)	0.235
	32	67 M	B	0.574 (0.009)	0.000 (0.000)	0.574
		67 M	W	0.241 (0.030)	0.000 (0.000)	0.241
	64	134 M	B	0.577 (0.008)	0.000 (0.000)	0.577
		134 M	W	0.251 (0.027)	0.000 (0.000)	0.251
	256	537 M	B	0.589 (0.016)	0.000 (0.000)	0.589
		537 M	W	0.323 (0.026)	0.000 (0.000)	0.323
	512	1074 M	B	0.668 (0.012)	0.000 (0.000)	0.668
		1074 M	W	0.482 (0.027)	0.000 (0.000)	0.482
	1000	2097 M	B	0.697 (0.005)	0.000 (0.000)	0.697
		2097 M	W	0.602 (0.014)	0.000 (0.000)	0.601
Pythia-1B	16	34 M	B	0.602 (0.005)	0.000 (0.000)	0.602
		34 M	W	0.263 (0.030)	0.000 (0.000)	0.262
	32	67 M	B	0.612 (0.006)	0.000 (0.000)	0.612
		67 M	W	0.268 (0.031)	0.000 (0.000)	0.268
	64	134 M	B	0.595 (0.014)	0.000 (0.000)	0.595
		134 M	W	0.232 (0.039)	0.000 (0.000)	0.232
	256	537 M	B	0.711 (0.014)	0.000 (0.000)	0.711
		537 M	W	0.473 (0.051)	0.000 (0.000)	0.473
	512	1074 M	B	0.695 (0.007)	0.000 (0.000)	0.695
		1074 M	W	0.509 (0.029)	0.000 (0.000)	0.509
	1000	2097 M	B	0.723 (0.004)	0.000 (0.000)	0.723
		2097 M	W	0.619 (0.019)	0.001 (0.000)	0.618